SCIENCE CHINA
Information Sciences

• REVIEW •

August 2026, Vol. 69, Iss. 8, 180301:1–180301:40
https://doi.org/10.1007/s11432-026-5040-3

Special Topic: Large Model-Driven Aerospace Microwave Imaging

# SAR foundation models: a comprehensive review of data, models, and applications

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Received 26 February 2026/Accepted 17 May 2026/Published online 27 July 2026

**Abstract** Synthetic aperture radar (SAR) provides all-weather, day-and-night observation for continuous global monitoring. The rise of foundation models has shifted artificial intelligence toward large-scale self-supervised pre-training followed by downstream adaptation. Transferring these models to the SAR domain, however, faces obstacles arising from complex microwave scattering mechanisms and the modality gap with optical imagery. Despite these challenges, a growing body of work on SAR foundation models has begun to emerge. To the best of our knowledge, this review presents the first unified taxonomy for SAR foundation models, systematically categorizing the landscape into visual, multimodal, and generative paradigms. From a data-centric perspective, we trace the transition from task-specific supervised learning to self-supervised pre-training and establish a taxonomy that organizes existing work into three paradigms: visual pre-training through masked image modeling and contrastive learning, multimodal foundation models encompassing SAR-optical synergy and vision-language interaction, and generative foundation models for data synthesis and cross-modal translation. We further review available datasets and evaluation benchmarks, and compare downstream adaptation strategies from full fine-tuning to parameter-efficient methods and prompt-based transfer. Finally, we analyze the main bottlenecks in this field and discuss future directions covering dataset construction, multimodal SAR agents, physical interpretability, and robustness under extreme imaging conditions.

**Keywords** SAR, foundation models, pre-training, multimodal learning, generative models

**Citation** Hou Y Y, Wu H R, Zhang Y L, et al. SAR foundation models: a comprehensive review of data, models, and applications. Sci China Inf Sci, 2026, 69(8): 180301, https://doi.org/10.1007/s11432-026-5040-3

## 1 Introduction

SAR, enabled by its active microwave imaging mechanism, overcomes the limitations imposed by illumination and weather conditions, allowing all-day and all-weather data acquisition. SAR has therefore become an important technology in domains such as disaster monitoring, resource exploration, and national defense. Early SAR image analysis commonly relied on handcrafted features and task-specific supervised pipelines; later, data-driven SAR approaches based on convolutional neural networks (CNNs) were explored for tasks such as automatic target recognition (ATR) [1–3]. These fully supervised methods tend to perform well under in-distribution settings with limited acquisition-condition variability, but with the rapid growth of data volume and increasingly complex interpretation requirements, their generalization limitations and annotation costs have become progressively more pronounced.

In recent years, foundation models built upon Transformer architectures have driven a paradigm shift in both natural language processing and computer vision [4–7]. Through large-scale self-supervised pre-training [8–13], these models have demonstrated remarkable general-purpose representation capabilities, exhibiting strong few-shot and even zero-shot transfer performance across a wide range of tasks. This trajectory, from task-specific training toward general representation learning, is relevant to SAR because it offers a way to exploit large volumes of unlabeled data and to reduce dependence on task-specific annotation. Its practical value, however, depends on whether the learned representations remain transferable across sensors, geometries, and downstream tasks. In this context, this survey focuses on SAR foundation models: general-purpose models pre-trained on large-scale unimodal SAR data

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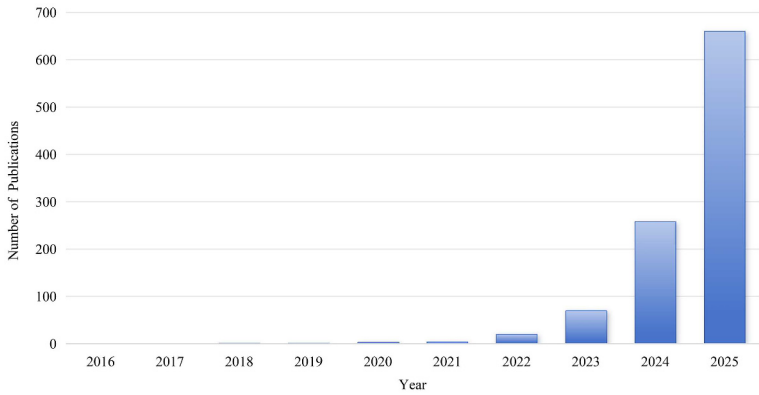



**Figure 1** (Color online) Annual number of publications related to SAR foundation models (2016–2025). The data are based on a Google Scholar search with the query: intitle: (SAR OR "synthetic aperture radar") AND ("foundation model" OR "large-scale pre-training"). Data retrieved on April 22, 2026.

or multimodal pairs (e.g., SAR-optical or SAR-text), whose core objective is to learn robust feature representations that encode intrinsic scattering mechanisms and can be adapted to diverse downstream tasks. A formal definition and the inclusion criteria used throughout the review are given in Subsection 3.1.

Transferring this paradigm to the SAR domain, however, is non-trivial and faces several distinctive obstacles.

(1) **Lack of SAR-tailored architectural designs.** Mainstream pre-training architectures (e.g., Vision Transformer (ViT) [14] and ResNet [15]) are largely designed around the visual statistics of optical imagery and do not explicitly account for SAR-specific imaging physics, particularly coherent scattering effects and complex-valued measurements. Directly applying these architectures may therefore fail to fully capture the diversity of electromagnetic scattering behaviors.

(2) **Generalization challenges induced by multi-dimensional imaging parameters.** SAR backscattering fundamentally depends on observation geometry and system configuration. Variations in frequency band, resolution, incidence angle, and polarization can produce substantial differences in SAR backscatter and image statistics, as discussed in general SAR imaging references and demonstrated in application-specific studies [16–18], which often degrade cross-condition transfer.

(3) **Difficulty in constructing large-scale, high-quality datasets.** Compared with optical imagery, SAR images are considerably harder to interpret visually, making annotation costly and largely dependent on scarce domain expertise [16]. Substantial heterogeneity across platforms and frequency bands further complicates data standardization.

Despite these challenges, research on SAR foundation models has grown rapidly in recent years [19,20]. As shown in Figure 1, the number of relevant publications has increased markedly since 2016. The field nonetheless remains at a fragmented, exploratory stage, and a systematic technical taxonomy with in-depth analysis of the key challenges is still lacking.

Several surveys on related topics have been published as interest in Earth observation foundation models has grown. Table 1 compares these studies with the scope of this review (✓: discussed; ✗: not discussed). Xiao et al. [21] and Huang et al. [22] provided high-level overviews of remote-sensing foundation models, highlighting the generalization benefits of large-scale pre-training on optical and multimodal data, but both treat SAR as a complementary modality and do not examine the implications of SAR's distinctive imaging geometry and electromagnetic scattering characteristics. Mai et al. [23] discussed foundation models from a broader geospatial artificial intelligence (GeoAI) perspective and covered knowledge graphs and multimodal reasoning, but overlooked the complex-valued and phase-related properties specific to SAR. Wang et al. [24] offered a systematic account of self-supervised representation learning in remote sensing; however, most reviewed methods rely on data augmentation strategies designed for optical imagery, such as color transformations, which may corrupt physically meaningful information when applied to SAR data. Li et al. [25] covered remote-sensing vision-language models in detail but are largely confined to discriminative tasks. Huang et al. [26] dedicated a review to generative AI in SAR yet focused primarily on traditional generative adversarial networks (GANs) and early diffusion-model applications, with limited treatment of large-scale pre-trained generative models and adaptation strategies. Miller et al. [27] approached satellite imagery from a time-series perspective, but did not explicitly discuss SAR-specific physical mechanisms, cross-sensor evaluation, or inference cost considerations. However, none of these studies provides a unified taxonomy of foundation models dedicated to SAR.

**Table 1** Comparison of this review with existing surveys and position papers. Unlike broader surveys or those focused on specific sub-tasks, this work provides a unified taxonomy dedicated to SAR foundation models and further incorporates practical discussions on SAR-specific physical mechanisms, data scale, cross-sensor evaluation, and inference cost. RS: remote sensing; SAR: dedicated to SAR; FMs: foundation models; SSL: self-supervised learning; VLMs: vision-language models; genAI: generative artificial intelligence; Phys.: physics mechanism; Vis.: visual representation; MM.: multimodal alignment; Gen.: generative diffusion; Eff.: efficient adaptation; Data: data scale analysis; X-sens.: cross-sensor evaluation; Cost: inference cost discussion.

| Reference | Core scope | SAR | Phys. | Vis. | MM. | Gen. | Eff. | Data | X-sens. | Cost |
|---|---|---|---|---|---|---|---|---|---|---|
| Xiao et al. [21] | General RS FMs | ✗ | ✗ | ✓ | ✓ | ✗ | ✓ | ✓ | ✗ | ✗ |
| Huang et al. [22] | General RS FMs | ✗ | ✗ | ✓ | ✓ | ✗ | ✓ | ✓ | ✗ | ✓ |
| Mai et al. [23] | GeoAI FMs | ✗ | ✗ | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Wang et al. [24] | RS SSL | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ | ✓ | ✗ | ✗ |
| Li et al. [25] | RS VLMs | ✗ | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Miller et al. [27] | RS time-series | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Huang et al. [26] | SAR genAI | ✓ | ✗ | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ |
| Ours | SAR FMs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

To bridge this gap, this work provides a comprehensive review of SAR foundation models. The main contributions are as follows.

(1) **Systematic taxonomy of SAR foundation models.** This work presents a comprehensive review of SAR foundation models. From a data-centric perspective, we trace the transition from task-specific supervised learning to large-scale self-supervised pre-training and establish a taxonomy that organizes existing work into visual, multimodal, and generative paradigms.

(2) **In-depth analysis of SAR-specific learning and adaptation mechanisms.** We examine how pre-training strategies such as masked image modeling (MIM) and contrastive learning are adapted to SAR-specific challenges including speckle noise and modality gaps, explore multimodal foundation models encompassing SAR-optical synergy and vision-language interaction, and discuss generative foundation models for data synthesis and cross-modal translation. We also summarize downstream adaptation strategies from full fine-tuning to parameter-efficient methods and prompt-based transfer.

(3) **Open challenges and future directions.** We analyze the main bottlenecks facing current SAR foundation models and discuss future research directions, covering large-scale dataset construction, multimodal SAR agent development, interpretability grounded in physical mechanisms, and robustness under extreme and complex imaging conditions.

As illustrated in Figure 2, the remainder of this survey is organized as follows. Section 2 provides background knowledge on foundation models and SAR imaging. Section 3 outlines the proposed taxonomy, which is subsequently detailed across three core categories: SAR visual foundation models via MIM and contrastive learning (Section 4), multimodal foundation models including vision-language and large language models (Section 5), and SAR generative models for text-to-SAR and image-to-SAR tasks (Section 6). Section 7 reviews pre-training datasets and evaluation benchmarks. Section 8 discusses downstream adaptation techniques, including fine-tuning and inference paradigms like zero-shot prompting. Finally, Section 9 outlines future directions, and Section 10 concludes the paper.

## 2 Background

### 2.1 SAR imaging mechanism and data-driven learning evolution

SAR forms high-resolution imagery by coherently integrating radar echoes along the flight path of a moving platform, synthesizing an aperture far larger than the physical antenna. Because the image arises from coherent superposition of backscattered microwave signals, its properties differ fundamentally from optical imagery. Speckle, caused by constructive and destructive interference within each resolution cell, introduces granular multiplicative noise that obscures edges and weak scatterers. The slant-range projection inherent to side-looking acquisition produces foreshortening, layover, and radar shadow, distorting spatial relationships in ways that have no counterpart in nadir-viewing optical data. Moreover, the raw SAR signal is complex-valued, with the phase component encoding path-length differences essential for interferometric and polarimetric analysis, yet discarded when only amplitude images are used. These properties establish a substantial domain gap with respect to optical imagery, limiting the direct transfer of models developed for natural images.

Before the introduction of deep learning, SAR interpretation relied on handcrafted feature descriptors, notably histogram of oriented gradients (HOG) and gray-level co-occurrence matrix statistics, fed into classifiers such as

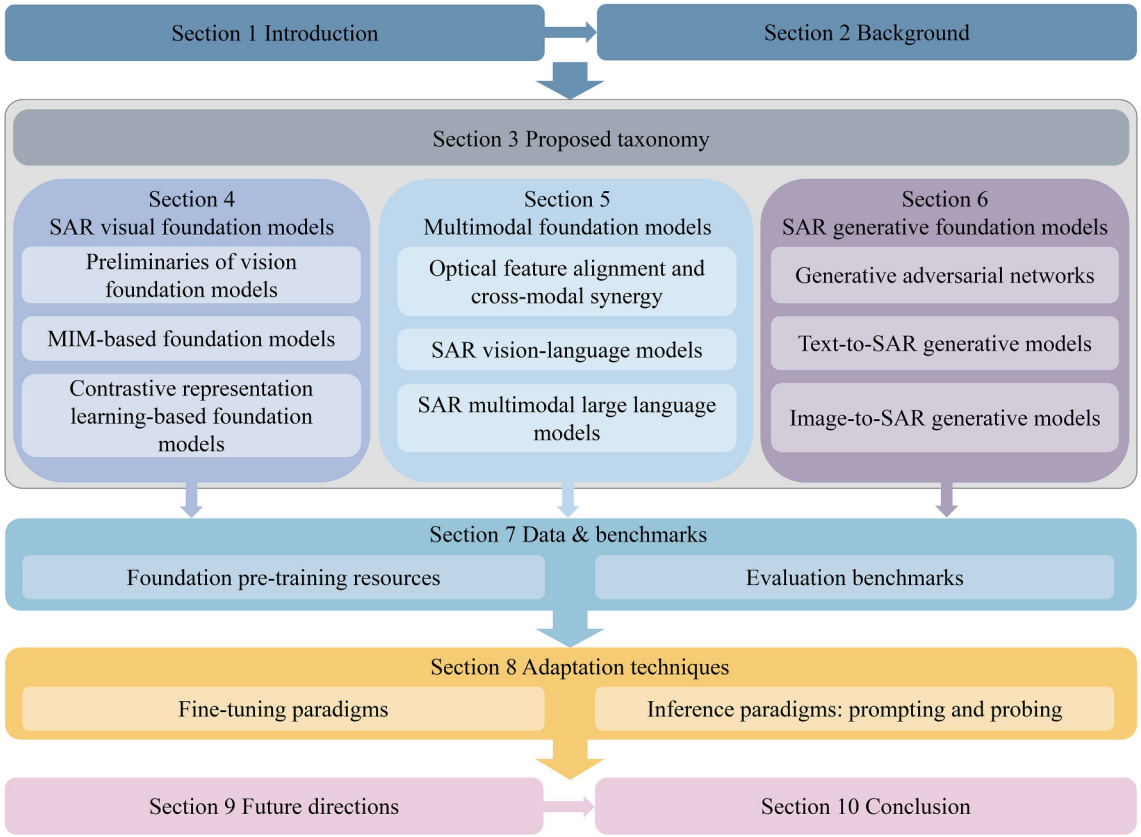



**Figure 2** (Color online) Organization of the survey.

support vector machines (SVMs). These pipelines demanded considerable domain expertise and their performance plateaued as scene complexity grew. Starting around 2015, CNNs entered the SAR domain through SAR ATR and detection studies [1–3], offering end-to-end learned feature hierarchies that rapidly became the prevailing approach across SAR interpretation tasks.

ATR, long a central problem in SAR, was among the first to benefit. Morgan [1] demonstrated competitive CNN-based ATR on MSTAR and analyzed adaptation to previously unseen target classes. Using the MSTAR benchmark, Chen et al. [2] showed that end-to-end convolutional architectures yield more discriminative representations than hybrid pipelines retaining handcrafted stages. Ding et al. [3] further investigated CNN training with data augmentation, including translation, speckle-noise injection, and pose-aspect variation, to improve recognition robustness under limited training data. Wagner [28] replaced the fully connected layers with an SVM to improve decision-boundary robustness.

Object detection in SAR poses further challenges from strong background clutter and arbitrary target orientation. Kang et al. [29] adapted Faster R-CNN with contextual information from surrounding sea-surface regions to suppress clutter-induced false alarms. Jiao et al. [30] developed a densely connected multiscale Faster R-CNN framework, in which multi-level feature fusion improves proposal generation and detection robustness for small ships in cluttered SAR scenes. To handle orientation, An et al. [31] developed an SAR-specific rotated bounding-box detector, while R3Det [32] provides a general rotated-object detection framework that can be transferred to remote-sensing targets. These oriented detectors better fit elongated or obliquely oriented objects and can reduce missed detections and localization error when adapted to SAR scenes.

For pixel-level interpretation, fully convolutional networks and encoder-decoder architectures such as U-Net have been applied to road extraction from SAR images [33], semantic segmentation of PolSAR land-cover scenes [34], and oil-slick monitoring from Sentinel-1 imagery [35]. A distinct challenge is the exploitation of polarimetric information: polarimetric SAR (PolSAR) data record the full scattering matrix, yet standard real-valued CNNs do not natively preserve complex-valued algebra, phase coupling, and Hermitian covariance structure. Zhang et al. [36] proposed complex-valued CNNs that operate directly in the complex domain, demonstrating that preserving phase through complex convolutions yields superior land-cover discrimination.

At the signal level, CNN-based despeckling has offered a compelling alternative to classical filters such as Lee and

Frost, which tend to over-smooth fine structures. Chierchia et al. [37] proposed SAR-CNN with a residual learning formulation that predicts the speckle component, preserving spatial detail while suppressing noise. Wang et al. [38] introduced ID-CNN, combining Euclidean and total-variation losses to improve perceptual quality. Since supervised despeckling requires speckle-free ground truth that is rarely available, Lattari et al. [39] adopted a modified U-Net encoder-decoder, first pre-training on synthetically generated speckled data and then fine-tuning on real SAR images with total-variation regularization to preserve spatial structures.

These task-specific CNN advances have raised the performance ceiling of SAR interpretation considerably. Yet the development model remains fragmented: each application (ATR, ship detection, land-cover segmentation, despeckling) requires its own architecture, labeled dataset, and training cycle. A ship detector trained on Sentinel-1 maritime scenes, for instance, offers little reusable knowledge for land-cover mapping from the same sensor. This one-model-per-task paradigm, compounded by the high cost of expert annotation, has motivated the search for more general and label-efficient learning frameworks, setting the stage for the foundation-model approaches discussed below.

To facilitate precise discussion, we clarify several related yet distinct geometric terms in SAR imaging. Observation geometry is the broadest concept and describes the overall spatial configuration between the sensor and the target. It is determined by factors such as the incidence angle, aspect angle, and slant-range distance. The incidence angle refers to the angle between the radar beam and the local vertical, or equivalently the normal to the reference ground plane at the target location. It is a key factor that influences backscatter intensity, shadow formation, and radiometric variation. The aspect angle describes the horizontal angular relationship between the platform flight direction and the target orientation. Its variation may expose different scattering centers in complex targets and is therefore important for target recognition. Imaging geometry is closely related to observation geometry but has a narrower meaning. It usually refers to the projection relationship from the three-dimensional scene to the two-dimensional SAR image plane. Typical effects include slant-range versus ground-range projection, layover, foreshortening, and shadowing. In this sense, imaging geometry can be regarded as a specific manifestation of observation geometry under a given sensor configuration.

### 2.2 The rise of foundation models

The term foundation model was introduced by Bommasani et al. [40] to denote a model pretrained on broad data at scale, typically through self-supervised learning, and subsequently adapted to diverse downstream tasks via fine-tuning or prompting. This pretrain-then-adapt paradigm originated in natural language processing: BERT [8] showed that bidirectional masked-language pre-training yields contextual representations transferable to question answering, sentiment analysis, and named-entity recognition, while the GPT series [9–12] demonstrated that autoregressive pre-training at increasing scale leads to emergent few-shot and zero-shot capabilities. In computer vision, masked autoencoders (MAE) [13] extended a similar self-supervised philosophy to images, achieving competitive performance with vision transformers pretrained by reconstructing randomly masked patches. These successes have since been adopted in the remote-sensing community, motivating the construction of domain-specific foundation models for Earth observation. More broadly, recent studies on domain-specific foundation models have emphasized that general-purpose models often need domain-aware data construction, knowledge injection, and efficient adaptation mechanisms before they can be reliably transferred to specialized application scenarios [41]. This observation is particularly relevant to SAR interpretation, where the imaging mechanism, scattering characteristics, and sensor configurations differ substantially from those of natural images.

The appeal of foundation models for SAR interpretation stems from a straightforward observation: satellites such as Sentinel-1 and Gaofen-3 continuously generate vast volumes of unlabeled imagery, while high-quality annotations remain scarce due to the specialized expertise required for SAR data interpretation (see Section 1, Challenge 3). Self-supervised pre-training objectives, including masked patch reconstruction and contrastive instance discrimination, offer a way to exploit this abundance by learning spatial structures and scattering patterns directly from raw data, substantially relaxing the annotation bottleneck. Beyond data efficiency, foundation models also hold the promise of learning transferable representations that can be seamlessly adapted to multiple downstream tasks (e.g., via prompt-based paradigms such as the segment anything model (SAM) [42] and SAR-oriented prompt methods such as RSPrompter [43]), potentially replacing the conventional paradigm in which each application demands a separately trained model. However, as discussed in Section 1, realizing this promise in the SAR domain is nontrivial: mainstream scalable backbones (e.g., ViT [14] and swin transformer [44]) carry optical-domain inductive biases that may not capture the full diversity of electromagnetic scattering behaviors, and the sensitivity of SAR signatures to imaging geometry and system configuration poses additional obstacles to cross-sensor generalization. How to preserve the scalability and versatility of foundation models while addressing these SAR-specific challenges

constitutes the central question driving the research surveyed in this review.

Throughout this survey, the foundation model is used as the default term. Pre-trained model refers only to a model that has undergone pre-training, without implying broad transferability, while large model is avoided except when scale itself is being emphasized, in which case we use large-scale foundation model. The formal SAR-specific definition and inclusion criteria are given in Subsection 3.1.

## 3 Proposed taxonomy

### 3.1 Definition and inclusion criteria

Before presenting the taxonomy, we formalize what qualifies as an SAR foundation model in this survey and specify the inclusion criteria used throughout the review.

**Definition.** Building on the concept introduced by Bommasani et al. [40], we define an SAR foundation model as a model that satisfies all of the following conditions: (i) it is pre-trained on large-scale data that includes SAR imagery, either as the sole modality (unimodal) or paired with auxiliary modalities such as optical images or text (multimodal); (ii) its pre-training objective is general-purpose representation learning rather than optimization for a single downstream task; and (iii) it provides empirical evidence of transferability or adaptability across multiple downstream tasks, evaluation settings, or generation conditions.

**Inclusion criteria.** We operationalize this definition through three axes.

• *Pretraining data scale and diversity.* The model must be pretrained on a dataset that is substantially larger than conventional task benchmarks. The dataset should also cover diversity in at least one acquisition or scene dimension, including geographic coverage, sensor type, frequency band, polarization mode, or spatial resolution. Models trained only on a single small benchmark, such as MSTAR for both pretraining and evaluation, are excluded.

• *Generality of pretraining objectives.* The training objective must support general use rather than direct optimization for a single downstream metric. It should learn transferable representations, general conditional generation, or both. Eligible objectives include MIM, contrastive instance discrimination, cross-modal alignment, and diffusion-based generative modeling. For discriminative models, generality refers to representations that can transfer across tasks. For generative models, it refers to a controllable data distribution that generalizes across target categories, scenes, or imaging conditions, rather than reproducing one narrow dataset. Fully supervised models trained only with task-specific labels are excluded, even when they use large architectures. Generative models tuned only to imitate one restricted data source are also excluded unless they show such controllability.

• *Transferability evidence.* The model must show effectiveness beyond its original training setting. For discriminative models, evidence may include adaptation to multiple downstream tasks, transfer across sensor configurations or imaging geometries, or competitive few-shot or zero-shot performance. For generative models, evidence may include controllable synthesis or restoration across multiple categories, sensors, or acquisition conditions. Examples include generation of unseen categories controlled by prompts or conditions, as well as despeckling or cross-modal translation that transfers beyond the training corpus.

Under these criteria, generic vision foundation models (e.g., a standard ViT pre-trained solely on ImageNet) are not included unless they have undergone SAR-domain pre-training or adaptation that itself satisfies the above conditions. Likewise, early task-specific transfer-learning approaches that initialize SAR classifiers or detectors from natural-image pre-trained CNNs and then fine-tune them on SAR task data [45–47] are discussed in Section 4 as important precursors but are not classified as SAR foundation models.

The taxonomy presented below organizes the SAR foundation-model landscape, including closely related precursor and baseline methods, into three paradigms based on their core functional objectives and data modalities. As shown in Figure 3 [19,48–92], we categorize SAR foundation models into three major groups: SAR visual foundation models, SAR multimodal foundation models, and SAR generative foundation models. This taxonomy covers the main directions of existing techniques and also reflects the field's technological evolution, progressing from unimodal feature representation to cross-modal semantic alignment and further to high-fidelity data generation.

### 3.2 SAR visual foundation models

SAR visual foundation models focus on unimodal representation learning. Their core objective is to extract generic visual features with strong robustness and generalization directly from large-scale unlabeled SAR imagery, without requiring manual annotations, thereby addressing the sample-scarcity challenge in SAR image interpretation. According to the design principles of self-supervised pre-training tasks, we divide these models into two categories.

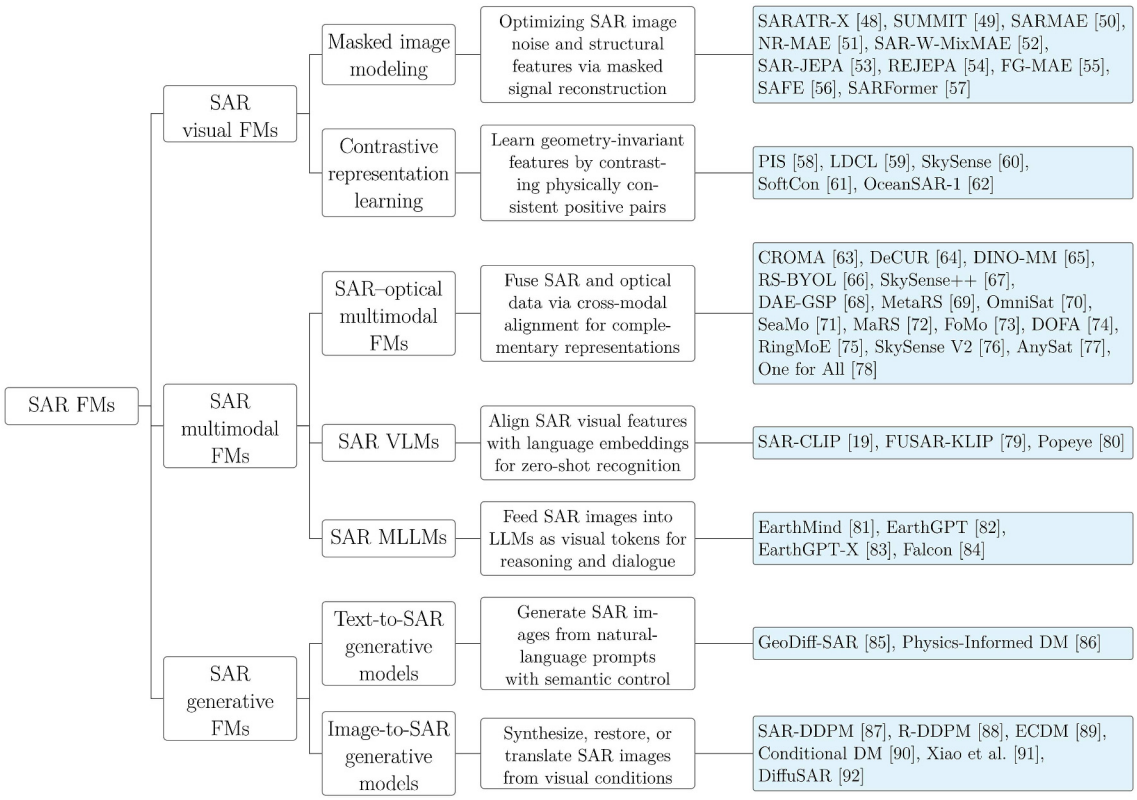



**Figure 3** (Color online) The proposed taxonomy of SAR foundation models.

(1) **MIM-based foundation models.** These methods emphasize reconstruction. By randomly masking parts of the input image (e.g., pixel patches or frequency-domain features), the model is forced to recover the missing signals using contextual information. This enables the model to capture the spatial dependencies and local texture structures within SAR imagery.

(2) **Contrastive representation learning-based foundation models.** These methods emphasize discrimination. By bringing homologous samples closer in the feature space and pushing heterogeneous samples farther apart, the model aims to learn globally discriminative features that are invariant to factors such as viewing geometry and noise.

### 3.3 SAR multimodal foundation models

SAR multimodal foundation models focus on cross-modal alignment and semantic reasoning. Given that SAR imagery lacks intuitive color and texture cues, these models introduce auxiliary modalities such as optical images or textual descriptions, aiming to bridge the semantic gap of radar-only data. We further divide this category into three types.

(1) **SAR-optical multimodal foundation models.** These models emphasize fusion at the physical-sensor level. They exploit the complementary nature of optical imagery and SAR, with a focus on feature registration across heterogeneous sensors and learning complementary representations.

(2) **SAR vision-language models.** These models emphasize semantic alignment between vision and language. By leveraging contrastive learning (e.g., the contrastive language-image pre-training paradigm), SAR visual features are mapped into a natural-language embedding space, enabling open-vocabulary recognition and zero-shot perception.

(3) **SAR multimodal large language models.** These models emphasize complex logical reasoning. They feed SAR images into large language models as visual embeddings and use instruction-following and language-mediated reasoning, subject to task-specific evaluation, to move beyond pure perception toward cognition, dialogue, and decision-making.

**Table 2** Reproducibility summary of representative SAR foundation models and closely related generative baselines. "Backbone" specifies the model version or main backbone used in the corresponding work. "Scale" reports the parameter count when it is available; otherwise, "Not reported" is used. In the "Code" column, "Released" indicates that official pretrained weights or checkpoints are publicly available or obtainable through the project's stated release channel. "Project page" indicates that a public project page is available, while the completeness of code or checkpoints should be verified.

| Paradigm | Model | Backbone | Scale | Code | Weights |
|---|---|---|---|---|---|
| MIM | SARATR-X [48] | HiViT-B | 66M | Released | Released |
| MIM | SUMMIT [49] | ViT-B | 86M | Released | Released |
| MIM | SARMAE [50] | ViT-B/L | 86M/307M | Released | Released |
| MIM | SAR-W-MixMAE [52] | Swin-B | 88M | Released | Released |
| MIM | SAR-JEPA [53] | ViT-B | 86M | Released | Released |
| MIM | FG-MAE [55] | ViT-S/B/L/H | 22M/86M/304M/658M | Released | Released |
| MIM | SAFE [56] | Masked Siamese ViTs | Not reported | Project page | Not reported |
| Contrastive | PIS [58] | ResNet-50/Swin-B | 23M/88M | Released | Released |
| Contrastive | SkySense [60] | Swin-H + ViT-L + Transformer | 2.06B | Released | Released |
| Contrastive | SoftCon [61] | RN50/ViT-S/ViT-B | 23M/23M/86M | Released | Released |
| Contrastive | OceanSAR-1 [62] | ResNet50/ViT-S/ViT-S/ViT-B | 23M/21M/21M/85M | Released | Released |
| SAR-optical | CROMA [63] | CROMA-B/L | 86M/307M | Released | Released |
| SAR-optical | DeCUR [64] | ResNet-50 | 23M | Released | Released |
| SAR-optical | DINO-MM [65] | ViT-S | 22M | Released | Released |
| SAR-optical | SkySense++ [67] | Swin-H + ViT-L + Transformer | 2B | Released | Released |
| SAR-optical | OmniSat [70] | OmniSat | 10M | Released | Released |
| SAR-optical | SeaMo [71] | ViT-B | 86M | Released | Released |
| SAR-optical | MaRS [72] | SwinV2-B | 88M | Released | Released |
| SAR-optical | FoMo [73] | FoMo-Net1/m | 66.39M–128.96M | Released | Released |
| SAR-optical | DOFA [74] | ViT-B/L | 86M/304M | Released | Released |
| SAR-optical | RingMoE [75] | RingMoE | 14.7B/6.5B/4.3B/1B | Released | Not reported |
| SAR-optical | AnySat [77] | AnySat | 125M | Released | Released |
| SAR VLM | SAR-CLIP [19] | AnyReader ViT-B/L + Transformer text encoder | 167.5M/463.5M | Released | Released |
| SAR VLM | FUSAR-KLIP [79] | ViT-B/L + BERT | 198M/420M | Released | Not reported |
| SAR MLLM | EarthGPT [82] | DINOv2 ViT-L/14 + CLIP ConvNeXt-L + LLaMA-2-7B | 9.2B | Project page | Not reported |
| SAR MLLM | EarthGPT-X [83] | DINOv2 ViT-L/14 + CLIP ConvNeXt-L LLaMA-2-13B + SAM | 19.5B | Project page | Not reported |
| SAR MLLM | Falcon [84] | Florence-2 + Transformer | 0.7B/0.3B | Released | Released |
| Text-to-SAR | GeoDiff-SAR [85] | SD3.5-M + LoRA + geometry-conditioned fusion | LoRA params not reported | Released | Released |
| Image-to-SAR | SAR-DDPM [87] | U-Net | Not reported | Released | Released |
| Image-to-SAR | DiffuSAR [92] | U-Net | 1.1M | Project page | Not reported |

### 3.4 SAR generative foundation models

SAR generative foundation models focus on data synthesis and physical enhancement. To address shortages of task-specific training data and annotations, as well as severe speckle corruption, these models leverage the powerful distribution-modeling capacity of diffusion models to synthesize supplementary data or restore degraded imagery. Within the scope of this survey, generative foundation models are organized according to their conditioning signals into two categories: text-to-SAR generative models and image-to-SAR generative models. This division reflects the two dominant routes by which current generative foundation models control SAR synthesis and restoration.

(1) **Text-to-SAR generative models.** These methods focus on semantically driven controllable synthesis. By incorporating pre-trained text encoders such as CLIP, the model can interpret natural-language prompts (e.g., "oil tankers in a harbor" or "vehicles on a snowy background") and generate SAR images with specified target categories and scene attributes, enabling open-vocabulary generation and zero-shot recognition.

(2) **Image-to-SAR generative models.** These methods focus on structurally driven generation and restoration. Using visual signals (e.g., layout masks, noisy images, optical base maps) as strong conditioning inputs, they encompass tasks ranging from layout-guided data synthesis and image despeckling to cross-modal translation, aiming to preserve physical details while augmenting training data.

Although this taxonomy focuses on diffusion-based foundation models, early GAN-based methods remain important technical precursors. They are therefore briefly reviewed at the beginning of Section 6 to provide historical context and to clarify how later diffusion-based models inherited and extended earlier SAR generative designs.

The taxonomy above clarifies the model families covered in this survey. For practical reuse, readers also need to know whether representative models are publicly reproducible and how large they are. Table 2 summarizes the backbone or model variant, parameter scale, public code availability, and pretrained-weight availability of representative SAR foundation models and closely related generative baselines.

As summarized in Table 3, the choice of model family and adaptation method is primarily driven by data scale and compute budget, with PEFT and few-shot probing offering effective alternatives when labeled data are scarce.

The taxonomy separates SAR foundation models by functional objective: visual representation learning, multimodal alignment and reasoning, and generative modeling. For reuse, model choice should be guided jointly by data modality, label budget, compute budget, and whether code and weights are available.

**Table 3** Practical decision guide for selecting SAR foundation model families and adaptation strategies under different input data, label availability, compute budget, and task requirements. Model families correspond to the taxonomy in Section 3; adaptation strategies are detailed in Section 8. FFT = full fine-tuning; PEFT = parameter-efficient fine-tuning; ZS = zero-shot; FS = few-shot probing; MIM = masked image modeling; CRL = contrastive representation learning; pol. = polarization.

| Input data | Labeled data | Compute | Target task | Model family | Adaptation | Representative methods/notes |
|---|---|---|---|---|---|---|
| **SAR-only input** | | | | | | |
| SAR (single-/dual-pol) | Large | High (GPU cluster) | Detection, segmentation, dense prediction | SAR visual FM | FFT | SARATR-X [48], SUMMIT [49]; two-stage SAR pre-training on large-scale unlabeled SAR corpora, such as SARDet-180K in SARATR-X [48], followed by task FFT. Progressive transfer eases domain shift. |
| SAR (single-/dual-pol) | Medium | Moderate (single GPU) | Classification, detection | SAR visual FM | PEFT: LoRA/adapter | SARMAE [50], NR-MAE [51] + LoRA on attention layers; ClassWise-SAM-Adapter [93]. Freezes backbone; injects SAR physical priors efficiently. |
| SAR (single-/dual-pol) | Small | Limited/edge | ATR, scene classification | SAR visual FM | FS probing | Freeze backbone; train linear head only, LDCL [59]. Metric learning. Avoids catastrophic forgetting; suitable for edge deployment. |
| SAR | ZS | Inference only | Open-vocabulary recognition, segmentation | SAR visual FM or SAR VLM | ZS prompting | SAR-CLIP [19] for text prompts; RSPrompter [43]/RingMo-SAM [94] for visual prompts into SAM. Avoids FFT of the backbone by using ZS text prompting or lightweight prompt/adaptation modules. |
| **Multimodal input** | | | | | | |
| SAR + optical | Medium-large | Moderate-high | Land-cover mapping, change detection, fusion | SAR-optical multimodal FM | PEFT or FFT | CROMA [63], DeCUR [64], DOFA [74]. PEFT preferred under scarce pairs; FFT when pairs are abundant. Random modality masking in DINO-MM [65] improves robustness to missing modalities. |
| SAR + text descriptions | Small-medium | Low | ZS recognition, open-vocabulary retrieval | SAR VLM | ZS prompting | FUSAR-KLIP [79]. Knowledge-guided contrastive learning injects geoscientific and SAR-domain priors to overcome the physics-semantics gap. |
| SAR + text | Medium | Moderate | VQA, dialogue, chain-of-thought (CoT) reasoning | SAR MLLM | PEFT: LoRA on interface | EarthGPT [82], EarthGPT-X [83], Falcon [84]. Typically freeze or largely preserve the LLM backbone while adapting the vision–language interface. Preserves LLM linguistic priors at low parameter cost. |
| Any sensor (arbitrary bands) | Medium | Moderate | Multi-sensor/any-modality tasks | Universal FM | PEFT or ZS | DOFA [74], AnySat [77], RingMoE [75]. Sensor-adaptive or modality-adaptive modeling across SAR, optical, multispectral, and other Earth observation inputs. |
| **Generative/data-scarcity scenarios** | | | | | | |
| SAR images | N/A (generative task) | Moderate | Data augmentation, despeckling, layout-guided synthesis | SAR generative FM image-to-SAR | FFT on U-Net + LoRA on encoder | SAR-DDPM [87], DiffuSAR [92], ECDM [89]. FFT on U-Net ensures geometric/physical fidelity; LoRA on text encoder preserves semantic controllability. Frequency-domain modules sharpen speckle statistics. |
| Natural-language prompts (no SAR input required) | N/A (generative task) | Moderate | Controllable SAR generation, ZS recognition via synthesis | SAR generative FM (text-to-SAR) | ZS generation | GeoDiff-SAR [85], Physics-informed DM [86]. SARCLIP-1M [19] and SAR-TEXT [95] provide image-text resources for semantic conditioning. Generated pseudo-samples can train classifiers for unseen categories. |

## 4 SAR visual foundation models

SAR image interpretation has long been confronted with two intertwined challenges: the scarcity of high-quality annotated data and the complexity of SAR imaging mechanisms, including speckle noise and geometric distortions. Conventional fully supervised learning paradigms rely heavily on costly manual annotations, and model generalization is often limited when dealing with the non-intuitive visual characteristics of SAR imagery. In recent years, research on SAR visual foundation models has increasingly drawn on SSL paradigms such as contrastive learning and MIM [13, 48, 49, 96, 97]. In this section, starting from the preliminaries of visual foundation models, we provide a systematic review of self-supervised training strategies, with a detailed taxonomy shown in Figure 4.

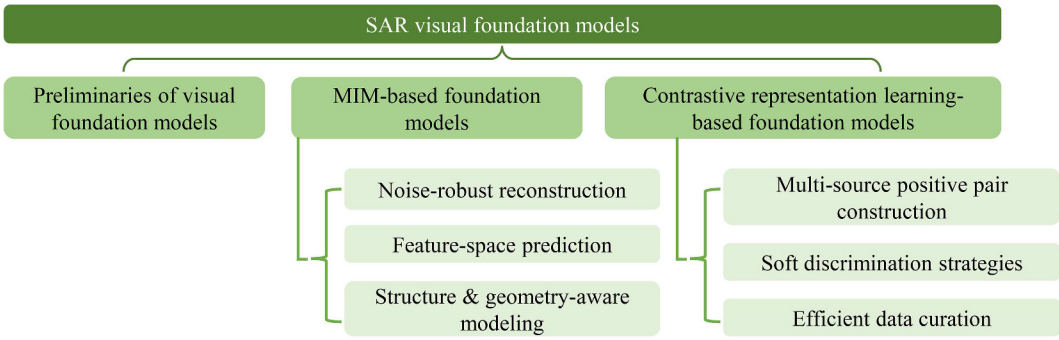



**Figure 4** (Color online) Taxonomy of enabling techniques for SAR visual foundation models.

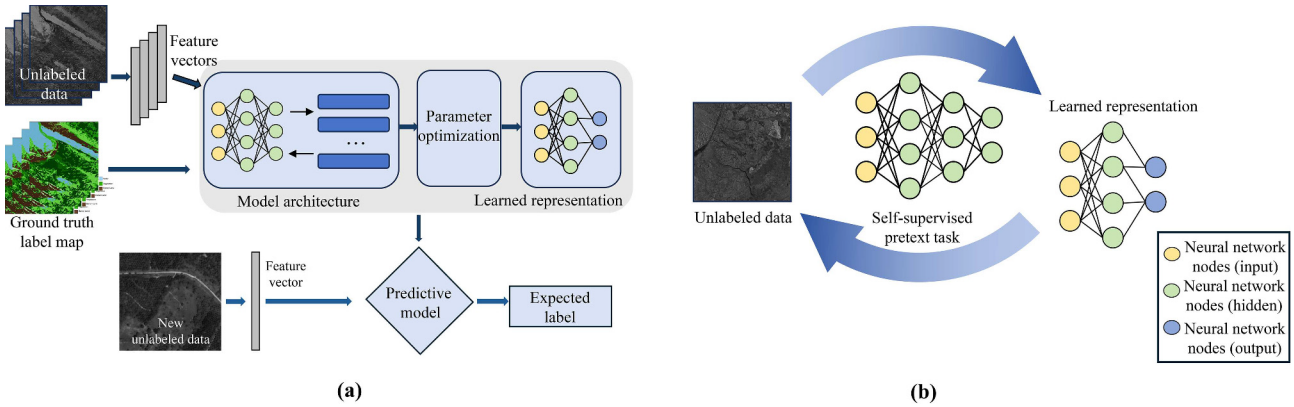



**Figure 5** (Color online) Comparison of two learning paradigms. (a) Supervised learning paradigm; (b) self-supervised learning paradigm.

### 4.1 Preliminaries of visual foundation models

In the early stage of SAR visual foundation model development, research primarily focused on transfer learning based on optical pre-trained weights [45,46]. However, due to the substantial modality discrepancy between optical and SAR imagery, directly transferring generic visual features often fails to accommodate SAR-specific scattering mechanisms [98]. Moreover, unlike natural-image recognition settings supported by ImageNet-scale labeled data [99], fully supervised SAR datasets are relatively limited in scale and diversity, making it difficult to train large-parameter SAR models entirely from scratch. To address this bottleneck, work such as SARATR-X [48] began to explore the potential of pre-training with large-scale unlabeled SAR data. As illustrated in Figure 5, this transition essentially relocates the source of supervisory signals from manual labels to the data itself. By exploiting the intrinsic structural patterns within massive unlabeled datasets to optimize model parameters, this approach is intended to learn representations less dependent on task-specific labels and more robust to SAR-domain variability [100]. Specifically, by first conducting masked pre-training on ImageNet to learn general shape priors and then performing a second-stage pre-training on the curated SARDet-180K pre-training corpus [48], this approach demonstrated that progressive self-supervised pre-training can yield more robust SAR visual representations than supervised transfer learning. To better handle the small-target setting, this work further adopted the HiViT [101] architecture, whose hierarchical MIM-compatible design improves efficiency and downstream transfer compared with a plain ViT [14].

Currently, pre-training for SAR visual foundation models is mainly built upon two self-supervised mechanisms: MIM and contrastive representation learning.

MIM focuses on generative learning and aims to explore the data distribution through reconstruction-based objectives [13, 102–104]. The core idea is to randomly mask local regions of an input image, thereby forcing the model to leverage contextual information to reconstruct the missing signals. Formally, given an input SAR image $x$, a set of masked regions $\mathcal{M}$ is randomly sampled according to the masking ratio. The pre-training objective is to minimize the reconstruction error:

$$\mathcal{L}_{\mathrm{MIM}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|x_i - \hat{x}_i\|_2^2, \tag{1}$$

where $x_i$ and $\hat{x}_i$ denote the original pixel/feature value and the predicted value of the $i$-th image patch, respectively.

Contrastive representation learning emphasizes instance discrimination [96,105–110]. This paradigm learns globally discriminative feature representations by pulling positive pairs closer in the feature space while pushing negative pairs farther apart. Its core objective typically optimizes the InfoNCE [111] loss:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\exp(\text{sim}(z_i, z_i^+)/\tau) + \sum_{j=1}^{K}\exp(\text{sim}(z_i, z_{i,j}^-)/\tau)}, \tag{2}$$

where $z_i$ denotes the query feature, $z_i^+$ denotes its positive feature, $z_{i,j}^-$ denotes the $j$-th negative feature for $z_i$, $K$ is the number of negatives, and $\tau$ is the temperature.

Although these general paradigms have achieved remarkable success in computer vision, directly applying them to SAR imagery still faces substantial challenges. The pervasive multiplicative speckle noise in SAR images can easily cause MIM-based models to overfit high-frequency noise rather than the underlying physical structures. Meanwhile, the strong sensitivity of SAR targets to viewing geometry makes the construction of positive pairs in contrastive learning particularly difficult. Therefore, the central focus of research on SAR visual foundation models lies in how to adapt these paradigms to the physical characteristics of SAR.

### 4.2 MIM-based foundation models

Naively applying pixel-level MIM to SAR imagery can be problematic because speckle and extreme backscatter values may dominate the reconstruction loss. If one blindly pursues pixel-level reconstruction as in (1), the model may overemphasize random noise or very bright scatterers rather than learning the underlying backscattering characteristics of ground targets. Consequently, the evolution of generative representation learning for SAR is essentially a shift from high-fidelity pixel recovery toward physical feature decoupling.

To avoid blindly fitting speckle noise, recent studies have begun to modify the reconstruction target of MIM so that it explicitly incorporates SAR physical priors. The common strategy underlying these methods is to transform the plain inpainting task into an enhancement-and-restoration task. SUMMIT [49] pretrains a ViT-MAE on 560000 SAR images. In addition to the classic MIM objective of reconstructing the original image, it introduces two in-domain auxiliary heads: a denoising head and an edge-scattering-point head. The denoising head adopts additive Gaussian noise as the degradation model, constraining the network to recover a clean SAR image under masking; the edge-scattering-point head uses Canny edges and Harris strong scattering points as pseudo-labels to supervise the reconstruction of high-frequency structures. By sharing the encoder and fusing the three reconstruction losses with learned weights, the model explicitly learns to suppress speckle while preserving edges and scattering points already during pre-training. This work shows that, even when relying entirely on SAR signals, MIM can still significantly improve the edge sharpness and semantic discriminability of SAR foundation models, as long as the auxiliary tasks are coupled with the imaging physics. Similarly, SARMAE [50] and NR-MAE [51] focus on noise modeling: the former enhances representation robustness by injecting synthetic speckle noise (the SARE module) and requires the model to reconstruct the original, uncorrupted signal, while the latter directly integrates a denoising task into the masked reconstruction pipeline. SAR-W-MixMAE [52] further introduces a backscatter-power-weighted reconstruction loss for MixMAE pre-training on Sentinel-1 imagery, reducing the influence of speckle and very high backscatter values during representation learning.

To fundamentally circumvent the sensitivity of pixel-level reconstruction to noise, some studies have shifted toward prediction in feature space. SAR-JEPA [53] proposes a model based on a joint-embedding predictive architecture (JEPA), abandoning the direct regression to the raw pixels $x_i$ in (1) and instead predicting the latent representations of the masked regions in feature space. By using multi-scale gradient features as guidance signals, SAR-JEPA steers the model to focus on the geometric contours of targets rather than background clutter, since gradient features mainly reflect the rate of intensity change and are naturally robust to multiplicative speckle noise. This formulation also brings significant gains in computational efficiency. Choudhury et al. proposed REJEPA [54], further validating the efficiency advantages of the JEPA framework for remote sensing image retrieval. REJEPA discards the computationally expensive decoder and predicts abstract representations of target regions in feature space using contextual information.

Beyond noise suppression and feature-space prediction, a third line of work pursues structure and geometry-aware pre-training, aiming to encode the spatial topology and imaging geometry of SAR scenes rather than pixel-level textures. In terms of structured-information reconstruction, FG-MAE [55] introduces the HOG as the reconstruction target, encouraging the model to recover edge- and corner-related structures of man-made targets such as bridges and buildings. SAFE [56] uses masked Siamese ViTs and SAR-specific augmentations to learn robust SAR feature extractors across acquisition modes and resolutions, rather than emphasizing pixel-wise reconstruction. Moving from

structural features to imaging geometry, SARFormer [57] conditions pre-training on physical acquisition parameters. To address the non-object-centered nature of remote sensing data, this work designs a keep-masking strategy to preserve the integrity of geographical observations and introduces an acquisition-parameter encoding (APE) module that embeds physical metadata into the tokens. This design is intended to make representations more sensitive to acquisition geometry and related SAR distortions. Collectively, these studies indicate that for SAR foundation models, learning structural and acquisition-aware context can be more useful for generalization than pursuing only fine-grained pixel recovery.

### 4.3 Contrastive representation learning-based foundation models

The key to applying contrastive learning in the SAR domain lies in constructing positive pairs that are consistent with physical principles. Although general methods such as SimCLR [96] and MoCo [106] have been introduced, conventional image augmentations (e.g., color transformations) often distort SAR amplitude and phase information and cannot emulate the complex variations induced by SAR imaging mechanisms. Benchmark studies such as PhilEO Bench [112] indicate that the generalization capability of most current models in complex scenes still needs to be improved.

Liu et al. [113] proposed a multi-source remote sensing contrastive self-supervised learning (CSSL) paradigm. Instead of being restricted to augmentations of a single image, this method leverages multimodal or multi-view data to construct a Siamese network, driving the model to learn modality-shared structural features that are robust to sensor parameters. An et al. [58] further exploited physical consistency along the temporal dimension and proposed a strategy for promoting intra-instance similarity (PIS). By using imagery of the same geographic location at different time phases to form positive samples, PIS encourages the model to reduce sensitivity to environmental disturbances such as seasonal land-cover changes and acquisition-condition variability, and to focus on more stable representations of ground objects. The success of this strategy indicates that spatio-temporal correlation serves as a natural form of augmentation and offers an effective pathway to building generalizable remote sensing foundation models.

Standard contrastive learning (as in (2)) assumes that all negative samples are mutually exclusive. This hard-discrimination assumption is particularly prone to failure in the SAR domain, because the same target observed from different aspect angles may appear drastically different, while targets from different categories may appear highly similar. To address this issue, Liao et al. [59] proposed low-confidence discrimination contrastive learning (LDCL). This study points out that the high inter-class similarity in SAR imagery can lead the model to overly confident erroneous gradients during instance discrimination. By introducing a group-instance discrimination mechanism and a batch-mixing strategy, LDCL proactively reduces the model's discrimination confidence for specific instances, thereby yielding a more inclusive clustered distribution in the feature space. SoftCon, proposed by Wang et al. [61], approaches the problem from the label perspective: it abandons the mutual-exclusivity constraint of one-hot encoding and introduces soft similarity constraints to allow reasonable semantic overlap in the feature space. To completely avoid the noise interference caused by negative-sample selection, Nawaz et al. [114] adopted a bootstrap your own latent (BYOL), training the network solely by maximizing feature consistency between augmented views. The evolution of this line of work reveals that, for SAR foundation models, appropriately introducing ambiguity and soft constraints can capture the intrinsic data structure better than forcing strong instance separability.

Beyond the design of views and learning mechanisms, handling the unique data redundancy of SAR imagery is also a key challenge. OceanSAR-1, proposed by Kerdreux et al. [62], shifts the focus to efficient data-level utilization to reduce redundancy. The study points out that SAR data for ocean monitoring often exhibit extreme redundancy and a long-tailed distribution, and that blindly scaling up the dataset size is not necessarily the optimal solution. To this end, OceanSAR-1 introduces a dynamic dataset curation strategy that, during DINO self-supervised training, automatically removes semantically repetitive high-frequency background samples such as sea-surface clutter. While significantly reducing pre-training computational cost, this strategy effectively mitigates overfitting and achieves performance superior to general-purpose foundation models on regression tasks such as sea-surface wind speed and wave-height inversion.

The research surveyed in this section converges on a consistent finding: pixel-level reconstruction objectives, when applied to SAR without modification, cause the model to fit multiplicative speckle rather than learn target structure. Redirecting the reconstruction target to feature space, as in SAR-JEPA [53], or to structural descriptors such as HOG, substantially reduces this risk and produces representations that transfer more reliably to downstream tasks. The two-stage strategy adopted by SARATR-X [48] provides further empirical evidence on this point. Generic optical pre-training serves primarily as parameter initialization; the performance gains critical to SAR tasks originate from the subsequent SAR-specific pre-training stage, where the model learns to disentangle scattering signatures

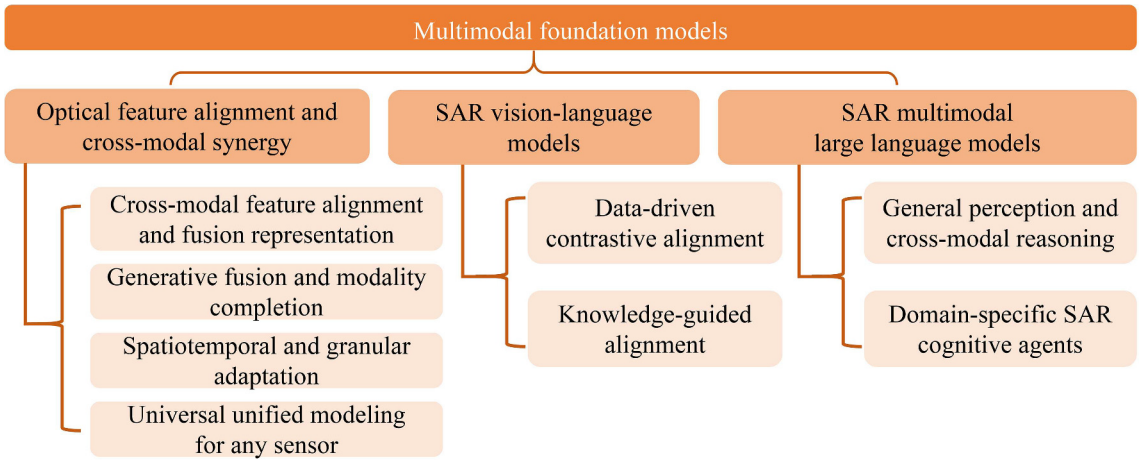



**Figure 6** (Color online) Taxonomy of enabling techniques for multimodal foundation models.

from speckle. SARATR-X validates this across five datasets and eight evaluation settings, suggesting that for tasks sensitive to SAR imaging physics, SAR-adapted pre-training is preferable to direct transfer of generic vision foundation models. A separate but equally important lesson emerges from OceanSAR-1 [62]: scaling dataset size alone does not guarantee improved representations when the data distribution is heavily long-tailed. Curating training samples to reduce semantic redundancy can be more effective than collecting additional data. Despite these advances, no unified benchmark currently exists for comparing SAR-specific pre-training against generic pre-training under controlled conditions across multiple tasks and sensors. The interplay between masking ratio, noise level, and downstream task type is also insufficiently understood, and most existing models have been validated on a narrow set of sensors, leaving cross-sensor generalization largely untested.

SAR visual foundation models benefit most when pre-training objectives suppress speckle-driven shortcuts and preserve structural scattering cues. Future comparisons should test SAR-specific and generic pre-training under controlled sensor, geometry, and task shifts.

## 5 Multimodal foundation models

Although SAR visual foundation models have achieved remarkable breakthroughs at the level of feature extraction, SAR imagery lacks color information and its textural characteristics deviate substantially from human visual experience, which causes purely visual models to face a severe interpretability bottleneck in semantic understanding. In contrast, optical images provide rich texture details consistent with human perception, while natural language highly abstracts semantic logic at a conceptual level. Therefore, establishing a cross-modal collaborative mechanism that introduces optical visual features or textual semantic priors into the SAR representation space is no longer merely an auxiliary approach, but a critical paradigm for breaking through the upper bound of SAR image interpretation capability. This section investigates how cross-modal interactions can bridge the gap between physical signals and high-level semantics. The detailed taxonomy is shown in Figure 6.

### 5.1 Optical feature alignment and cross-modal synergy

Optical and SAR images exhibit an inherent complementarity in describing ground objects: the former provides spectral reflectance and surface texture, while the latter reveals dielectric properties and geometric structures. The core challenge in this field lies in how to bridge the significant physical modality gap. Before foundation-model-based SAR-optical learning, task-specific studies mainly relied on homogeneous feature transformation or image-translation mechanisms to bridge heterogeneous optical and SAR observations, for example in optical-SAR change detection [115] and optical-to-SAR temporal shifting [116]. These approaches can reduce low-level modality gaps but may also weaken SAR-specific scattering cues when the alignment is forced too aggressively. In task-specific domains, cross-modal alignment between optical and SAR imagery has become a notable trend [117]; in the foundation-model domain, modern SAR-optical collaborative learning paradigms are evolving from simple feature pulling-together toward feature decoupling and complementarity.

#### 5.1.1 *Cross-modal feature alignment and fusion representation*

The primary challenge in heterogeneous sensor alignment is how to preserve modality-specific physical attributes while learning shared semantic commonalities. CROMA [63] separately encodes paired SAR and optical observations and combines cross-modal contrastive learning with masked reconstruction through a fusion encoder. Rather than using the optical branch as a one-way semantic teacher, it learns shared radar-optical representations while retaining modality-specific information through reconstruction objectives. This design acknowledges inter-modality differences, thereby preventing semantic collapse of SAR features during the process of moving toward the optical modality. However, soft alignment alone may be insufficient. DeCUR [64] introduces a feature decomposition paradigm that separates common and unique multimodal representations. By explicitly modeling both shared and modality-specific information, this decoupling mechanism helps integrate complementary radar-optical cues and improves robustness in modality-missing settings.

The random modality masking strategy in DINO-MM [65] provides a practical way to improve robustness to missing modalities. By training with optical-only, SAR-only, and SAR-optical inputs, the model is discouraged from relying on a single sensor stream and learns more robust joint representations. RS-BYOL [66] further extends this idea to multispectral and SAR data, demonstrating the potential of joint contrastive learning for extracting cross-sensor invariances.

Whether these strategies prevent modality collapse should be assessed explicitly. The main risk comes from initialization asymmetry: optical encoders are often pretrained on large natural-image corpora, whereas SAR encoders are trained from scratch or from smaller SAR datasets. As a result, the optical branch may dominate the shared feature space, leaving the SAR branch to act as a passive supplement rather than a self-sufficient representation of scattering structure.

A practical diagnostic is SAR-only evaluation after multimodal training. If disabling the optical branch causes a large performance drop, the SAR encoder has not learned robust standalone features. CROMA and DeCUR both report SAR-only linear probing results on BigEarthNet-MM, and their SAR branches retain meaningful discriminative capability. Most other SAR-optical methods still report only fused bimodal scores. We therefore recommend SAR-only testing as a routine diagnostic for distinguishing genuine cross-modal complementarity from optical-dominated fusion.

#### 5.1.2 *Generative fusion and modality completion*

With the rise of MIM, cross-modal fusion in remote sensing has developed along two related routes: explicit modality completion and representation-level multimodal learning. SkySense++ [67] follows the latter route as a semantic-enhanced multimodal remote-sensing foundation model. It adopts a factorized architecture and progressive pre-training on large-scale multimodal remote-sensing images, using multi-granularity contrastive learning for representation alignment and masked semantic learning for semantic enhancement. This training scheme strengthens multimodal representation learning and few-shot Earth-observation transfer by combining cross-modal alignment with semantic-level masking, without relying on pixel-level SAR-optical reconstruction.

DAE-GSP [68] further illustrates the value of representation-level modeling. It proposes a discriminative autoencoder with a Gaussian selective patch strategy for multimodal remote-sensing image classification. In this setting, Gaussian selective patching improves feature discrimination for multimodal classification and complements broader representation-level fusion strategies.

Although explicit pixel-level completion can be useful, it often introduces high computational cost and a risk of spurious textures or artifacts. Its robustness to missing modalities also requires careful validation. To address this limitation, MetaRS [69] explores a non-generative route. Rather than pursuing pixel-level cross-modal reconstruction, it designs a meta-modal consistency loss. The core idea is that, regardless of whether the input is unimodal or multimodal, the distribution of high-level features extracted by the model should remain consistent. The value of this strategy lies in achieving a modality-completion effect in feature space, improving feature-level consistency and robustness when optical inputs are missing (e.g., due to cloud or fog occlusion), without incurring the heavy computational overhead of pixel-level generation. Astruc et al. [70] also proposed the OmniSat architecture. This method maximizes mutual information among different modalities via SSL, improving missing-modality robustness within the evaluated remote-sensing tasks rather than guaranteeing complete scene understanding from unimodal input.

#### 5.1.3 *Spatiotemporal and granular adaptation*

Beyond modality differences, the temporal evolution and multi-scale characteristics inherent to remote-sensing data are also key issues that foundation models must address. SeaMo [71] addresses the value of temporal information for cross-modal alignment. Unlike static fusion, SeaMo introduces a temporal-multimodal fusion module that explicitly aligns optical and SAR imagery across different seasons, forcing the model to learn an essential land-surface representation that is robust to seasonal variations. This work successfully reframes the modality gap as part of broader spatiotemporal dynamics, achieving a higher-dimensional form of unification.

To address granularity mismatch in very-high-resolution (VHR) multimodal scenarios, MaRS [72] constructs the MaRS-16M dataset and proposes cross-granularity meta-modality learning. Its core components are cross-granularity contrastive learning (CGCL), which alleviates alignment inconsistency across paired samples, and meta-modality attention (MMA), which integrates heterogeneous modality characteristics. These components position MaRS as a granularity-aware SAR-optical representation-learning framework for VHR multimodal data. In addition, FoMo [73] further demonstrates the generalization advantages of such multi-source collaboration in vertical domains such as forest monitoring by constructing a multimodal, multi-scale benchmark covering satellite, airborne, and LiDAR data, providing strong evidence that multi-source collaborative sensing offers significant advantages over unimodal models for interpreting complex natural scenes such as biomass inversion.

#### 5.1.4 *Universal unified modeling for any sensor*

An ideal remote-sensing foundation model should be capable of handling inputs from arbitrary sensors, rather than being limited to specific modality combinations. Under this vision, Xiong et al. [74] proposed dynamic one-for-all (DOFA), which designs a wavelength-adaptive weight generation mechanism. Unlike approaches that simply treat SAR as a single-channel grayscale image, DOFA uses wavelength as a physical condition and feeds it into a hypernetwork to generate encoder parameters adapted to the current sensor. This mechanism allows the same Transformer backbone to process SAR data with parameter-level adaptation tailored to the SAR imaging mechanism. For large-scale multimodal visual interpretation, RingMoE [75] and SkySense V2 [76] adopt mixture-of-experts (MoE) designs to route heterogeneous remote-sensing modalities through specialized experts, supporting general-purpose remote-sensing representation learning and interpretation.

Building on this line of work, recent studies have begun to push the boundary of any-sensor modeling. Panopticon, proposed by Waldmann et al. [118], is built upon the DINOv2 framework and introduces channel subsampling and cross-channel attention mechanisms. This method treats observations of the same geographic location from different sensors as a natural form of data augmentation, thereby endowing the model with the flexibility to process inputs from arbitrary spectral bands and modalities. Similarly, AnySat proposed by Astruc et al. [77] combines a JEPA with a scale-adaptive encoder, aiming to address the high heterogeneity of remote-sensing data in terms of resolution, scale, and modality. Through unified training on the GeoPlex collection, it enables a monolithic model to cover multiple resolutions and land-surface monitoring tasks. One for all [78] is also devoted to building a unified Earth vision model, signaling that remote-sensing foundation models are moving from cross-domain adaptation toward multimodal unification. From different dimensions, including multi-sensor data augmentation and adaptation to resolution and scale heterogeneity, these studies collectively promote any-sensor modeling while avoiding an unsupported extension of spectral sensor-agnostic models to SAR bands or polarizations.

### 5.2 SAR vision-language models

The rise of VLMs [119–122] has promoted deep integration between vision and language. However, the pronounced non-intuitiveness of SAR imagery, including speckle noise, geometric distortions, and the absence of color and texture, makes semantic interpretation far more complex than for optical images. To bridge the gap between these physical characteristics and semantic descriptions, SAR vision-language models have emerged. As illustrated in Figure 7, such models mostly adopt a parallel dual-stream design: the visual branch extracts physical features, and the text branch extracts semantic features. The key lies in leveraging a multimodal aggregation module to precisely align abstract linguistic semantics with concrete SAR visual representations.

Due to the unique physical properties of SAR imagery, the commonly used strategy in optical remote sensing of reusing pretrained weights from natural images often fails, which has prompted researchers to redesign vision-language interaction mechanisms. Most SAR vision-language models draw on the original CLIP paradigm [123] and its remote-sensing adaptations [124, 125], aiming to map heterogeneous SAR visual features and text embeddings into a shared latent space via contrastive learning. According to whether external domain knowledge is incorporated into the alignment process, current representative studies can be broadly organized into two lines.

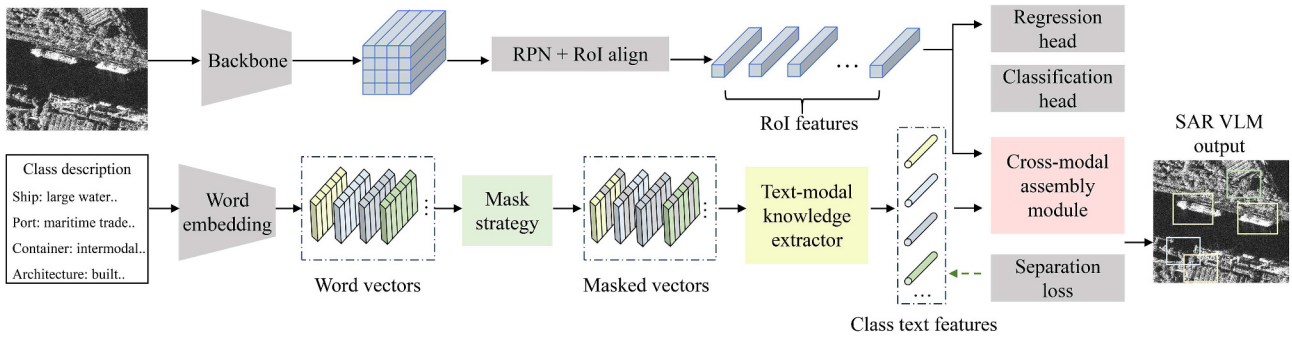



**Figure 7** (Color online) Schematic illustration of the multimodal model framework. This figure shows how the model aligns and fuses the visual features of SAR images with the textual semantic features of category descriptions via a multimodal module. RPN: region proposal network; RoI: region of interest.

#### 5.2.1 *Data-driven contrastive alignment*

SAR-CLIP [19] is one of the foundational studies in this area. Built upon the large-scale contrastive image-text pre-training paradigm, its core lies in exploiting paired SAR image-text data to learn cross-modal correspondences. By maximizing the cosine similarity of matched SAR image-text pairs and minimizing the similarity of mismatched pairs, SAR-CLIP learns cross-modal representations for retrieval and zero-shot recognition. Because the alignment is learned from data, the quality and diversity of the training corpus directly determine the semantic coverage of the resulting representation space. This dependence on data scale explains the construction of large SAR image-text resources such as SARCLIP-1M [19] and SAR-TEXT [95] discussed in Section 7.

#### 5.2.2 *Knowledge-guided alignment*

Purely data-driven contrastive pre-training can align SAR images with free-form text at the statistical level, but it has limited ability to capture domain-specific physical concepts that rarely appear in natural-language captions. Building on this direction, FUSAR-KLIP [79] introduces a knowledge-guided multimodal SAR foundation model. It constructs a large georeferenced SAR vision-language corpus with geographic projection attributes, generates structured SAR text through hierarchical cognitive thought chains, and uses a self-consistent iterative optimization mechanism combining contrastive, matching, and reconstruction objectives to improve SAR-text alignment under geoscientific and physical priors.

### 5.3 SAR multimodal large language models

Another frontier in the interdisciplinary field of remote sensing and artificial intelligence is the development of SAR-inclusive multimodal large language models (MLLMs) [81]. Unlike VLMs (e.g., SAR-CLIP) that focus on feature alignment, SAR-relevant MLLMs feed remote-sensing visual information into large language models, leveraging the LLM's generalization and reasoning capabilities to support interactive interpretation, visual question answering, and reasoning over target attributes or scene context. This architectural evolution indicates that SAR interpretation is shifting from task-specific pipelines driven by low-level features toward a general-purpose paradigm grounded in high-level semantics. The core idea is image-as-a-token, namely, converting SAR imagery into visual embeddings that the LLM can process through the following key components.

(1) **SAR image encoder.** This component encodes high-dimensional SAR pixel data into compact and semantically rich visual vectors. It typically adopts a pretrained SAR foundation model (e.g., SARMAE, SUMMIT) or a fine-tuned Vision Transformer, aiming to extract visual features useful for reasoning while preserving, as much as possible, physical information such as scattering mechanisms and geometric structures.

(2) **Vision-language interface.** Serving as a bridge between the visual encoder and the LLM, this module (e.g., a linear projector, Q-Former [120]) maps SAR visual tokens into the LLM's text-token space, enabling seamless cross-modal fusion. Depending on the adaptation design, the LLM may be frozen or partially fine-tuned; freezing the LLM and training only the interface layer is a common parameter-efficient choice that reduces computational cost and preserves linguistic priors.

(3) **Large language model.** Acting as the core reasoning engine, the LLM takes the fused multimodal tokens together with user textual prompts, leverages its CoT capability to parse complex contexts, and generates natural-language descriptions, reasoning results, or question-answering responses about the SAR image content.

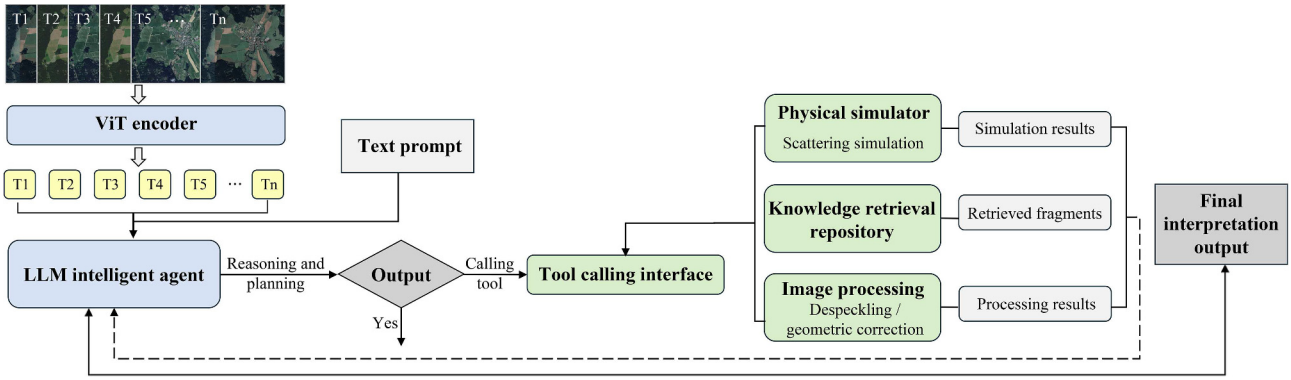



**Figure 8** (Color online) SAR cognitive agent workflow.

### 5.3.1 *General perception and cross-modal reasoning*

Traditional SAR interpretation typically relies on specialized models to handle classification, detection, or segmentation tasks separately, lacking holistic scene understanding and reasoning capability. The new generation of SAR-MLLMs breaks this task barrier by building a unified architecture for perception and reasoning.

EarthGPT [82] targets universal multisensor remote-sensing image comprehension by connecting remote-sensing visual representations with an LLM-based reasoning interface. EarthGPT-X [83] further extends this direction to multi-source imagery, including optical, SAR, and infrared data, and unifies referring and grounding tasks through visual prompting for spatial understanding. Falcon [84] establishes a unified prompt-based task paradigm, integrating remote-sensing tasks such as image classification, object detection, and semantic segmentation within a shared architecture.

General remote-sensing multimodal perception provides useful context for this transition. SkySense [60] unifies multi-source Earth-observation data within a foundation-model framework, offering broad multimodal representations for optical, SAR, and other remote-sensing inputs. Reframing SAR [126] complements model-centric studies by introducing a visual-reasoning dataset and analysis protocol for SAR target recognition, where multimodal LLMs are used to generate and evaluate CoT reasoning. Together, these studies broaden SAR interpretation from visual recognition toward multimodal perception and reasoning.

### 5.3.2 *Domain-specific SAR cognitive agents*

Although general-purpose foundation models are highly knowledgeable, they often lack sufficient domain depth when handling SAR-specific fine-grained tasks such as ship-type identification and marine environment analysis. Therefore, domain-specialized SAR models tailored to specific military or industrial applications have emerged. Figure 8 illustrates the typical workflow of such an SAR cognitive agent.

Popeye [80] proposes a unified visual-language model for multi-source ship detection, marking the evolution of vertical-domain interpretation from closed-set detection toward open-ended interactive localization. This model reduces dependence on specific sensors or annotation formats by designing a unified labeling paradigm. It reformulates coordinate regression as sequence generation, thereby unifying multi-source imagery and diverse detection tasks within a single visual-language framework. RingMoAgent [127] extends remote-sensing multimodal reasoning across optical, SAR, and infrared imagery from satellite and UAV platforms through the RS-VL3M vision-language dataset, modality-adaptive embeddings, task-specific tokens, and high-dimensional hidden-state decoding. REMSA [128] addresses model deployment by selecting suitable remote-sensing foundation models under task and resource constraints. These studies together show how SAR-inclusive interpretation is moving from closed-set recognition toward interactive localization, multimodal reasoning, and model-selection support.

Moving SAR-inclusive reasoning systems toward practical use requires addressing several engineering challenges that current work has only begun to expose. The first is context-length pressure. A single high-resolution SAR scene tokenized through a ViT encoder can produce thousands of visual tokens. When the task involves multi-temporal monitoring, the agent must reason over a sequence of acquisitions, and the cumulative token count can quickly exceed the context window of current LLMs. The second challenge is metadata reliability in agent workflows. Coordinate systems, incidence angles, orbit direction, polarization, and sensor-specific acquisition settings must be parsed correctly, otherwise errors can propagate through the reasoning chain. The third is inference latency. Disaster response and maritime surveillance demand near-real-time decisions, while autoregressive decoding accumulates

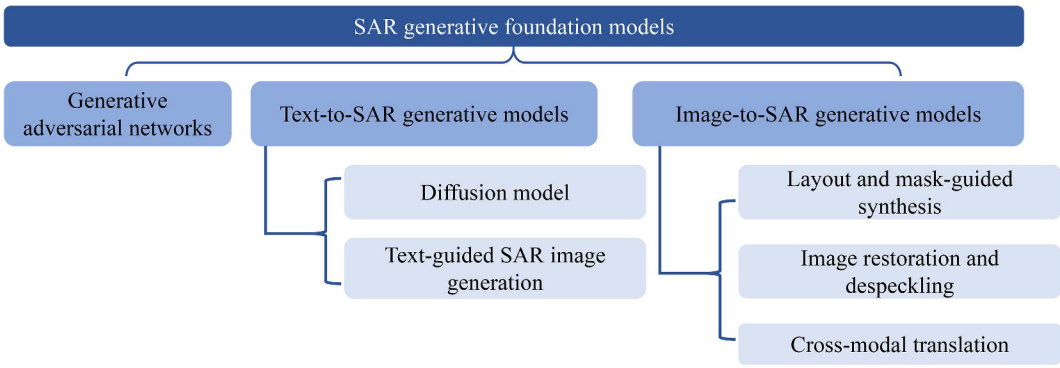



**Figure 9** (Color online) Taxonomy of enabling techniques for SAR generative foundation models.

delay across reasoning steps. Resolving these constraints is important for transitioning SAR-inclusive multimodal systems from laboratory demonstrations to operational use.

Drawing on the evidence presented throughout this section, several conclusions can be stated with reasonable confidence. SAR-optical complementarity is not merely a qualitative observation; quantitative results confirm that joint representations consistently outperform SAR-only representations. Under low-annotation regimes, CROMA [63] reports a 6.1 mean average precision (mAP) improvement for radar-optical over radar-only on BigEarthNet at 1% labels, and DeCUR [64] reports gains of 5.4 mAP in linear probing and 5.5 mAP in fine-tuning under the same label fraction. These gains indicate that optical features supply semantic context that SAR representations alone do not capture, and that this complementarity is most pronounced when labeled data are scarce. At the same time, the benefit is not unconditional. CROMA's ablation study shows that cross-modal patch-level InfoNCE fails to improve representation quality while slowing pre-training by 2.2×, and that hard-negative mixing degrades performance. DeCUR further demonstrates that removing intra-modal enhancement and feature decoupling reduces downstream SAR-optical performance by 1.4 mAP, confirming that preserving modality-unique information is essential and that forcing all features into a shared subspace suppresses the complementary advantage. For vision-language and agent-based models, the key open problems are grounding textual descriptions in SAR scattering semantics rather than visual appearance, and resolving the engineering constraints of context length, metadata reliability, and inference latency discussed above.

Multimodal SAR models are most useful when they preserve SAR-specific evidence while borrowing optical or language semantics. The main risk is over-alignment, where optical or textual priors dominate the radar branch and weaken physical grounding.

## 6 SAR generative foundation models

The core objectives of SAR image processing include image denoising, detail enhancement, and target recognition. However, issues arising from SAR imaging mechanisms, such as speckle noise, geometric distortions, and the scarcity of real measured data, severely limit further performance improvements of image processing algorithms. As an important branch of data-driven approaches, generative models offer new solutions to these challenges in SAR image processing. GANs [129] were once a major research focus in the generative modeling community, but they suffer from inherent drawbacks such as training instability and mode collapse. In recent years, diffusion models [130] have increasingly become the mainstream direction in generative modeling due to their stable training procedure, high generation quality, and strong capacity for prior representation, and they have demonstrated significant potential for applications in SAR image processing.

To systematically summarize the development trajectory of generative models for SAR, we divide the existing technical evolution into three stages: (i) early adversarial architectures that established the first data-driven generative paradigm for SAR imagery, (ii) text-guided generation with semantic control built upon diffusion models, and (iii) image-guided generation driven by visual structural cues. In particular, the mathematical preliminaries of diffusion models are introduced at the beginning of Subsection 6.2 to provide the necessary background for the subsequent discussion. This section reviews these stages in sequence (see Figure 9).

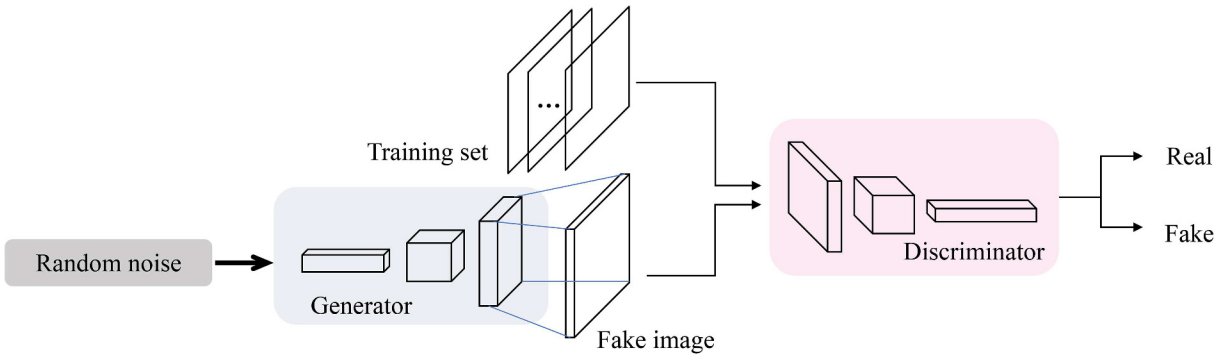



**Figure 10** (Color online) Schematic illustration of the basic architecture of GANs.

### 6.1 Generative adversarial networks

As the representative methodology in early generative modeling, GANs [129] (as shown in Figure 10) rely on an adversarial training paradigm between a generator and a discriminator. The GAN-based methods discussed in this subsection are mostly task-specific and therefore do not meet the inclusion criteria in Subsection 3.1. We review them here because they laid the foundation for SAR generative modeling, and several of their designs, such as multi-discriminator architectures, conditional generation modules, and physics-informed losses, have influenced later diffusion-based generative foundation models. While the adversarial mechanism allows GANs to rapidly learn core data features, direct application to SAR imagery presents unique challenges. SAR images are characterized by strong coherence, inherent speckle noise, and complex scattering properties. Consequently, traditional GAN architectures often struggle to capture the spatial structural rules and fine-grained target details of SAR scenes, which can lead to blurred target contours and logical inconsistencies in the generated scenes.

To mitigate these limitations, researchers have reconstructed GAN architectures and integrated multi-functional components to achieve refined control over the generation pipeline. Guo et al. [131] pioneered the application of GANs to SAR image generation by proposing an end-to-end synthetic model that inputs observation parameters (e.g., azimuth and depression angles) into a deconvolutional generator. Ju et al. [132] developed SARGAN, which integrates a scene constructor, target discriminators, and a generator to produce high-fidelity SAR scenes containing ships, land, and background elements. Furthermore, Oghim et al. [133] designed a dual-discriminator architecture specifically focused on high-frequency components, compelling the generator to simulate authentic high-frequency scattering characteristics and thereby enhancing structural details.

Beyond visual quality, downstream tasks heavily rely on controllable generation. To address category confusion and the lack of diversity caused by SAR data scarcity, researchers have introduced multi-dimensional constraints. Du et al. [134] incorporated category labels and random noise into the generator to improve sample diversity, while Oh et al. [135] proposed PeaceGAN, which adds a pose estimator and an auxiliary classifier to the discriminator to jointly train pose estimation, category recognition, and image generation. Other approaches, such as those by Wang et al. [136], Yang et al. [137], and Cui et al. [138], have successfully utilized angle-aware modules to achieve precise, continuous azimuth interpolation and controllable generation.

Finally, recognizing that SAR imaging is an active microwave process strictly governed by electromagnetic scattering laws, pure data-driven GANs risk generating distorted samples that violate physical principles. To address this, Zhang et al. [139] embedded a physical heuristic module based on the ideal point scattering center model into the network. By designing specific physical loss functions, they ensured that the generated images adhere to electromagnetic scattering laws while preventing discriminator overfitting in few-shot scenarios. Despite these advances, the adversarial training paradigm imposes a fundamental ceiling: GANs rely on a delicate balance between the generator and discriminator, making them prone to training instability and mode collapse, which limits their ability to capture the diverse scattering characteristics present in real SAR scenes. These limitations motivated the transition toward diffusion-based generative models, which are discussed in the remainder of this section.

### 6.2 Text-to-SAR generative models

#### 6.2.1 *Diffusion model*

To overcome the training instability and mode collapse inherent in adversarial methods, diffusion models adopt a fundamentally different generative paradigm. Rather than learning the data distribution implicitly through a

discriminator, they model the distribution through an iterative noising and denoising Markov process. This process consists of two phases: the forward diffusion process and the reverse denoising process.

In the forward diffusion process, the model incrementally injects Gaussian noise into the real SAR image $x_0$ over a series of steps, governed by a fixed variance schedule $\beta_t$. At time step $t$, the noised image $x_t$ is formulated as

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t\mathbf{I}). \tag{3}$$

As $t$ approaches the maximum step $T$, the original SAR signal is progressively destroyed, and the image degrades into isotropic Gaussian noise, yielding a well-defined prior distribution $x_T \sim \mathcal{N}(0, \mathbf{I})$.

In the reverse denoising process, a neural network $\epsilon_\theta(x_t, t)$ (typically based on U-Net) learns to invert this trajectory by predicting the injected noise at each step, thereby recovering the SAR image distribution from random noise. The training objective reduces to minimizing the mean squared error between the predicted noise and the ground-truth noise $\epsilon$:

$$\mathcal{L}_{\text{Diff}} = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\|_2^2\right], \tag{4}$$

where $\bar{\alpha}_t$ denotes the cumulative product of the noise-schedule parameters.

The decomposition of generation into denoising steps leads to a stable training process. Several architectures have been proposed to improve efficiency: denoising diffusion probabilistic models (DDPM) [130] established the theoretical framework; denoising diffusion implicit models (DDIM) [140] accelerated inference through non-Markovian sampling; and latent diffusion models (LDM) [141] projected the diffusion process into a compressed latent space to reduce computational cost. These properties make diffusion models well-suited for SAR image processing. Building on this foundation, the following subsection discusses how text-guided conditioning mechanisms are integrated into the diffusion framework to achieve semantically controllable SAR image generation; the same diffusion backbone also underpins the image-guided methods reviewed in Subsection 6.3.

#### 6.2.2 *Text-guided SAR image generation*

With the advance of vision-language pre-training, generating SAR images with specific attributes via natural-language prompts has emerged as a cutting-edge research topic. The core of these methods lies in building a cross-modal semantic interaction mechanism: they typically employ pre-trained text encoders such as CLIP or BERT to extract semantic embeddings, and inject them into the denoising network of a diffusion model via cross-attention, thereby enabling fine-grained control over SAR image texture, target categories, and background environments.

Large-scale SAR image-text resources such as SARCLIP-1M [19] and SAR-TEXT [95] provide paired semantic supervision for controllable SAR generation. Recent text- or parameter-conditioned generators, such as Physics-Informed DM [86] and GeoDiff-SAR [85], instantiate the generative side by injecting textual, physical, or geometric conditions into diffusion models.

Generative and vision-language models have also been used for zero-shot SAR target recognition through simulation-based data construction. Wang et al. [142] extracted target semantic information from optical remote-sensing images, transformed it into 3D target models for SAR simulation, and then reduced the simulated-to-measured domain gap with domain adaptation. The resulting framework couples optical semantic extraction, 3D target modeling, SAR simulation, and domain adaptation for zero-shot recognition.

In addition, to compensate for the limited physical fidelity of purely data-driven generation, some studies have explored physics-aware text-guided generation. Kamirul et al. [86] trained a diffusion model on physics-simulator-generated SAR wake images paired with text prompts derived from simulation parameters. This setting allows users to specify sea-state levels and vessel parameters through prompts, supporting fast and controllable Kelvin-wake image generation.

Despite their promising semantic controllability, current text-to-SAR generative models exhibit three recurring failure modes.

- **Semantics-physics mismatch.** A model may generate the requested object or scene category, such as a ship or harbor, while failing to reproduce the corresponding scattering pattern. This is especially problematic for targets defined by strong scatterers, shadow geometry, or multi-bounce interactions.
- **Prompt underspecification.** Natural-language prompts usually describe category and context, but rarely specify incidence angle, frequency band, polarization, or look number. The same prompt can therefore correspond to multiple physically distinct SAR observations.
- **Complex-scene hallucination.** In dense urban areas, rough sea surfaces, or mixed land-water boundaries, generators may smooth or hallucinate scattering structures that text alone cannot constrain.

These limitations indicate that text-to-SAR generation remains stronger in semantic plausibility than in physical faithfulness. Generated samples can support data augmentation or concept-level visualization, but they should be used cautiously in applications requiring reliable scattering statistics or radiometric interpretation.

### 6.3 Image-to-SAR generative models

Unlike text-to-image generation, which focuses primarily on semantic content, image-guided generation (Image-to-SAR) emphasizes using visual signals (e.g., layout masks, noisy images, optical images) as conditions to control the spatial structure and physical details of the generated images. Depending on the type of input condition, this section covers techniques including layout-guided synthesis, image inpainting and despeckling, and cross-modal translation.

#### 6.3.1 *Layout and mask-guided synthesis*

To generate SAR images with specific spatial distributions for alleviating data scarcity, researchers have proposed a layout-based generation paradigm. By imposing multi-dimensional constraints, these methods improve the accuracy and usability of generated data, enabling an integrated generation-annotation pipeline. Zhang et al. [143] introduced a multi-conditional DDPM that uses visual instances and environmental prompts as generation constraints, producing backscattering intensity information with fine-grained control at each resolution cell. Such constraint-driven generation can not only inpaint target regions in SAR images based on masks, but also output instance-level annotations simultaneously. Han et al. [144] generated high-quality SAR target images by conditioning on bounding boxes, providing an efficient data supplementation scheme for few-shot SAR target detection. To address data scarcity in specific scenarios, Jones et al. [145] proposed a generative-AI-based data augmentation framework. By generating synthetic training samples with precise annotations and diverse backgrounds, their approach improves detector performance on underrepresented SAR target classes and demonstrates the value of synthetic data as a complement to real training samples.

#### 6.3.2 *Image restoration and despeckling*

Image restoration and despeckling are essentially a generative process that maps degraded inputs to high-quality outputs. To account for the multiplicative nature of speckle noise in SAR imagery, researchers have introduced physics-level modifications to standard diffusion models.

For speckle-noise modeling, Perera et al. [87] introduced a DDPM-based SAR despeckling method, which iteratively restores clean images via a conditional noise predictor. Subsequently, to improve inference efficiency and detail preservation, Paul et al. [146] introduced an aggregated U-Net that fuses multi-stage denoising samples; Guo et al. [89] proposed an efficient conditional diffusion model (ECDM) that integrates a cosine noise schedule and a joint variance prediction mechanism; and Hu et al. [88] developed a region-wise denoising diffusion probabilistic model to accommodate large-scale images. For unsupervised settings, Xiao et al. [91] and Wang et al. [147] achieved self-supervised despeckling without paired data via a log-domain transformation and a diffusion posterior sampling framework, respectively.

Beyond statistical noise modeling, SAR-specific diffusion studies have begun to incorporate multi-scale or frequency-domain cues. Zhang and Li [90] proposed a conditional diffusion model for arbitrary-size SAR despeckling with multi-scale and dual-attention mechanisms. Guo et al. [89] further integrated wavelet transforms into ECDM to enhance structural fidelity across frequency bands. Ying et al. [92] explored frequency-aware SAR generation. As shown in Figure 11, these studies illustrate how frequency-domain priors can be incorporated into SAR-oriented generative restoration and enhancement.

#### 6.3.3 *Cross-modal translation*

To address the limited realism of certain data sources (e.g., simulated data), cross-modal translation has become an important means of expanding data provenance. Debuysère et al. [148] translated spaceborne SAR imagery into an airborne SAR representation, while also improving the realism of simulated images generated by the ONERA physical simulator, thereby bridging the gap between SAR data acquired from different platforms. Debuysère et al. [149] achieved high-fidelity SAR image generation through fine-tuning, which can be applied to resolution enhancement for TerraSAR-X satellite images and fidelity optimization for physics-simulator outputs, providing an effective solution for remote-sensing data augmentation and unseen-scenario simulation.

In summary, SAR generative models have evolved from early adversarial architectures to diffusion-based frameworks, advancing along two main axes: text-to-SAR synthesis for semantically controllable data augmentation, and

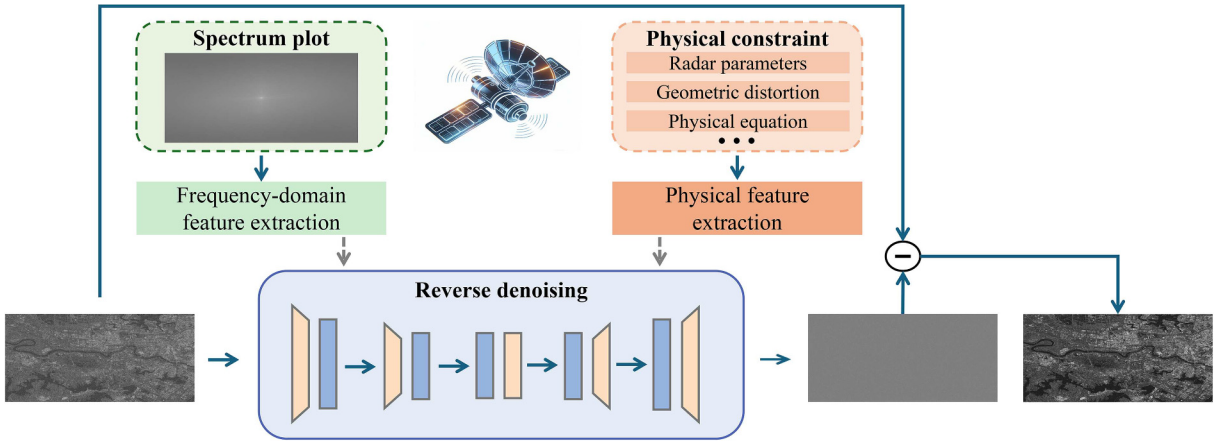



**Figure 11** (Color online) Diffusion models for SAR image restoration: incorporating frequency-domain priors (left branch) or integrating physical measurement constraints (right branch) into the reverse denoising process.

image-to-SAR generation for physical restoration and cross-modal translation. A recurring finding across despeckling and super-resolution methods is that incorporating SAR measurement models as explicit constraints within the diffusion process yields outputs whose intensity statistics are closer to real acquisitions than those produced by unconstrained generation. Yet current physical constraints remain limited in scope: only recently have geometric factors begun to be used as conditioning signals, and the systematic treatment of acquisition geometry, foreshortening, layover, and radiometric consistency remains limited. For text-to-SAR models, whether the generated images reproduce physically plausible scattering patterns rather than visually plausible textures has not been systematically tested, because mainstream evaluation metrics such as Fréchet inception distance (FID) and CLIP Score are agnostic to SAR imaging physics. Cross-modal translation methods have shown promise in bridging platform gaps, but their fidelity under large resolution or geometry differences between source and target sensors remains an open question. Developing evaluation protocols that go beyond perceptual similarity to quantify scattering-statistics consistency and radiometric fidelity, as discussed in Section 9, is a prerequisite for reliably assessing progress in this paradigm.

Generative SAR models are moving from visually plausible synthesis toward physically constrained generation. Their progress should be judged by scattering statistics, radiometric consistency, and downstream utility, not only by optical-domain perceptual metrics.

## 7 Data & benchmarks

The development of SAR foundation models relies on two types of data: large-scale unlabeled datasets for self-supervised pre-training, and annotated benchmarks for downstream adaptation and evaluation. Although algorithmic innovations often receive more attention, the generalization capability of these models largely depends on the scale, diversity, and quality of the available data. We categorize existing SAR data resources into two groups: pre-training datasets that support the construction of foundation models, and evaluation benchmarks used to assess task-specific performance. Figure 12 illustrates the mainstream data modalities in current research, ranging from uni-modal SAR to cross-modal pairings (e.g., SAR-optical and SAR-text), while Table 4 [19,20,48,50,55,95,99,150–180] provides a summary of the corresponding datasets, with public access links provided where available.

### 7.1 Foundation pre-training resources

According to differences in pre-training paradigms, we divide data resources into two categories: large-scale unsupervised or weakly supervised data for training robust vision backbones, and data designed for multimodal alignment.

#### 7.1.1 *Datasets for SAR visual foundation models*

The continuous provision of all-weather, global-coverage data by satellites such as Sentinel-1 has enabled the creation of million-scale unsupervised SAR pre-training corpora. A prominent example is SSL4EO-S12 [178], which offers millions of globally sampled SAR patches, effectively alleviating data scarcity for training parameter-heavy vision backbones. To further advance representation learning for high-resolution and complex scenarios, SARDet-100K

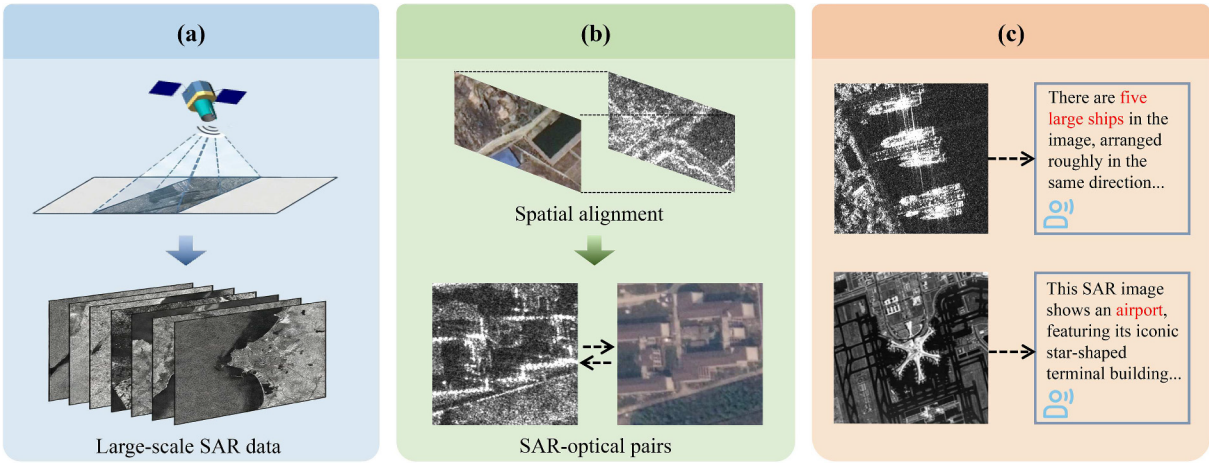



**Figure 12** (Color online) Illustration of mainstream data modalities in SAR foundation model research. (a) Uni-modal SAR: large-scale SAR data acquired directly from sensors for representation learning; (b) multi-sensor data: spatially registered SAR-optical image pairs for cross-modal learning; (c) vision-language pairs: SAR image-text pairs for semantic alignment.

[176] integrates multi-source data and establishes a benchmark at the scale of hundreds of thousands of images. For domain-specific representation, datasets such as FUSAR-Ship [161] provide extensive maritime repositories. Such domain-specific datasets are valuable for specialized pre-training because they contain dense coverage of particular application scenarios. Their transfer value outside these domains, however, can be limited by narrow sensor diversity and scene scope.

### 7.1.2 *Datasets for SAR multimodal foundation models*

The integration of multimodal information aims to overcome the inherent challenges of SAR imagery, specifically its sparse semantic details and difficulty in interpretation. We categorize existing multimodal resources into three groups: SAR-optical datasets, SAR vision-language datasets, and datasets for instruction fine-tuning and inference.

**SAR-optical datasets.** To leverage the rich spectral and texture details of optical imagery for better SAR interpretation and generation, creating pixel-level aligned SAR-optical datasets is essential. Early datasets, such as SEN1-2 [154] and OSdataset [160], mainly focused on precise geometric alignment, laying the foundation for data fusion. To increase geographic and climatic diversity, SEN12MS [156] and BigEarthNet-MM [165] provided hundreds of thousands of multimodal triplets covering various global climates and landscapes. These resources allow models to learn the physical connections between different sensors. Recently, the focus has shifted toward semantic consistency. Datasets like QXS-SAROPT [166] provide aligned SAR-optical image pairs for fusion tasks such as image matching and optical-information-assisted ship detection, while WHU-SAR-OPT [172] supports joint optical-SAR semantic segmentation and land-use classification. By aligning homogeneous ground objects across the two modalities, these datasets help models learn associations between SAR scattering features and optical spectral and textural information, effectively bridging the representation gap between different sensors.

**SAR vision-language datasets.** Constructing large-scale image-text pairs is essential for open-world SAR understanding and for providing semantic conditions to downstream generative studies. Generating textual descriptions for SAR imagery requires substantial domain expertise, resulting in a scarcity of high-quality paired data. To address this bottleneck, resources such as SARCLIP-1M [19] and SAR-TEXT [95] aggregate or construct paired SAR image-text data through dataset collection and annotation-to-caption pipelines. These resources support SAR representation learning, retrieval, captioning, and prompt-conditioned generation research.

**Instruction tuning and inference.** With the rise of multimodal large language models, the demand for data has shifted from simple image-text matching toward more complex logical reasoning and conversational interaction. Addressing this need, SARChat-2M [20] constructs a large SAR image-text/dialogue benchmark for evaluating VLM-based SAR interpretation, including visual understanding and object-detection-oriented tasks. Meanwhile, FSAR-Cap [180] employs a two-stage annotation strategy to build a captioning dataset with manually refined semantic descriptions. These resources expand SAR evaluation from image-text matching toward captioning, dialogue-style interpretation, and task-oriented multimodal understanding.

**Table 4** Summary of representative SAR datasets and benchmarks. Public access links are provided where available. Plat.: platform (Sat: satellite, Air: airborne, Multi: multiple); Res.: spatial resolution; Pol.: polarization; Var.: variable; Class.: classification; Obj. Det.: object detection; Seg.: segmentation; Cap.: captioning; Chat: visual question answering; Reg.: registration; V-L: vision-language. Count follows each dataset's native unit (e.g., images, patches, pairs, objects, samples, or dialogues).

| Dataset | Task | Plat. | Year | Count | Size | Sensor | Res. (m) | Pol. |
|---|---|---|---|---|---|---|---|---|
| MSTAR [150] | Class. | Air | 1996 | 14557 | Var. | STARLOS | 0.3 | HH |
| SSDD [151] | Obj. Det. | Sat | 2017 | 1160 | Var. | RadarSat-2/Sentinel-1/TerraSAR-X | 1–15 | HH/VV/HV/VH |
| OpenSARShip [152] | Obj. Det. | Sat | 2017 | 11346 | Var. | Sentinel-1 | 2.7–22 | VV/VH |
| OpenSARShip 2.0 [153] | Obj. Det. | Sat | 2017 | 34528 | 120 × 120 | Sentinel-1 | 10 | VV/VH |
| SEN1-2 [154] | Fusion/Matching. | Sat | 2018 | 282384 | 256 × 256 | Sentinel-1/Sentinel-2 | 10 | VV/VH |
| SAR-Ship-Det [155] | Obj. Det. | Sat | 2019 | 43819 | 256 × 256 | GF-3/Sentinel-1 | 1.7–25 | HH/VV/HV/VH |
| SEN12MS [156] | Class./Seg. | Sat | 2019 | 180662 | 256 × 256 | Sentinel-1/Sentinel-2 | 10 | VV/VH |
| AIR-SARShip-1.0 [157] | Obj. Det. | Sat | 2019 | 31 | 3000 × 3000 | GF-3 | 1–3 | VV |
| AIR-SARShip-2 [157] | Obj. Det. | Sat | 2020 | 300 | 1000 × 1000 | GF-3 | 1–3 | VV |
| LS-SSDD [158] | Obj. Det. | Sat | 2020 | 9000 | 800 × 800 | Sentinel-1 | 20 | VV/VH |
| HRSID [159] | Obj. Det. | Sat | 2020 | 5604 | 800 × 800 | Sentinel-1/TerraSAR-X | 0.5–3 | HH/VV/HV/VH |
| OSdataset [160] | Reg. | Sat | 2020 | 13365 | Var. | GF-3 | 1 | HH/VV/HV/VH |
| FUSAR-Ship [161] | Obj. Det./Class. | Sat | 2020 | 16144 | 512 × 512 | GF-3 | 1.7 | HH/HV |
| xView3-SAR [162] | Obj. Det. | Sat | 2021 | 991 | Var. | Sentinel-1 | 10 | VV/VH |
| SRSDD-v1.0 [163] | Obj. Det. | Sat | 2021 | 666 | 1024 × 1024 | GF-3 | 1 | HH/VV |
| ETCI2021 [164] | Seg. | Sat | 2021 | 66000 | 256 × 256 | Sentinel-1 | 10 | VV/VH |
| BigEarthNet-MM [165] | Class. | Sat | 2021 | 590326 | 120 × 120 | Sentinel-1/Sentinel-2 | 10 | VV/VH |
| QXS-SAROPT [166] | Obj. Det. | Sat | 2021 | 20000 | 256 × 256 | GF-3/Google Earth optical | 1 | HH |
| SI-STSAR-7 [167] | Class. | Sat | 2022 | 5915 | 256 × 256 | Sentinel-1 time-series SAR | 1–20 | HH/HV |
| MSAR [168] | Obj. Det. | Sat | 2022 | 28449 | 256 × 256 | GF-3/HISEA-1 | 1–5 | HH/VV/HV/VH |
| SADD [169] | Obj. Det. | Sat | 2022 | 2966 | 224 × 224 | TerraSAR-X | 0.5–3 | HH |
| RSDD-SAR [170] | Obj. Det. | Sat | 2022 | 7000 | 512 × 512 | GF-3/TerraSAR-X | 1–3 | HH/VV/HV |
| MMFlood [171] | Seg. | Sat | 2022 | 1748 | 2000 × 2000 | Sentinel-1 | 10 | VV/VH |
| WHU-SAR-OPT [172] | Seg./Class. | Sat | 2022 | 100 | 5556 × 3704 | GF-3 | 5 | HH/VV |
| SAR-AIRcraft [173] | Obj. Det. | Sat | 2023 | 4368 | Var. | GF-3 | 1 | HH/VV |
| OGSOD [174] | Obj. Det. | Sat | 2023 | 18331 | 256 × 256 | GF-3 | 3 | VV/VH |
| SIVED [175] | Obj. Det. | Air | 2023 | 1044 | 512 × 512 | Airborne | 0.1 | HH/VV |
| EuroSAT-SAR [55] | Class. | Sat | 2024 | 27000 | 64 × 64 | Sentinel-1 | 10 | VV/VH |
| SARDet-100K [176] | Obj. Det. | Sat | 2024 | 116598 | 512 × 512 | GF-3/Sentinel-1 | 1–20 | HH/VV/HV/VH |
| SARDet-180K [48] | Pre-train | Multi | 2024 | ∼0.18M | Var. | Multi | 0.1–25 | HH/VV/HV/VH |
| FAIR-CSAR [177] | Obj. Det. | Sat | 2025 | 340000 | Var. | GF-3 | 1–5 | HH/VV/HV/VH |
| SAR-TEXT [95] | Cap. | Sat | 2025 | 130000 | Var. | Sentinel-1 | 10 | VV/VH |
| SARCLIP-1M [19] | V-L pre-train | Multi | 2025 | >1M | Var. | Multi | Var. | Var. |
| SARChat-2M [20] | Chat | Sat | 2025 | 2000000 | 256 × 256 | Sentinel-1 | 10 | VV/VH |
| SSL4EO-S12 [178] | Pre-train | Sat | 2025 | 246000 | 264 × 264 | Multi | 10 | VV/VH |
| ATRNet-STAR [179] | Class. | Air | 2025 | 194324 | Var. | Airborne | 0.15 | HH/VV/HV/VH |
| SAR-1M [50] | Pre-train | Multi | 2025 | 1300000 | Var. | Multi | 0.1–60 | HH/VV/HV/VH |
| FSAR-Cap [180] | Cap. | Sat | 2025 | 14480 | Var. | GF-3 | 1–5 | HH/VV/HV/VH |

### 7.1.3 *Impact of dataset dimensions on generalization*

Beyond sheer volume, the physical acquisition parameters of pre-training data fundamentally constrain the generalization boundary of SAR foundation models. As discussed in Section 2, SAR backscattering is governed by sensor configuration, and variations in any single imaging dimension can alter the statistical properties of the same ground target [17, 18].

Frequency band controls penetration depth and dominant scattering mechanisms. C-band signals interact primarily with the canopy surface, whereas L-band signals penetrate into the vegetation volume, producing fundamentally different backscatter signatures [16]. A model pre-trained exclusively on C-band data therefore encodes surface-scattering priors that are unlikely to transfer to L-band foliage-penetration tasks. Even between bands of closer wavelength, non-trivial discrepancies arise: X-band sensors such as TerraSAR-X operate at shorter wavelengths than C-band Sentinel-1, yielding higher sensitivity to small-scale surface roughness and stronger specular returns from planar structures [16]. Features learned from medium-resolution C-band imagery may therefore misrepresent the

scattering behavior of the same targets observed at X-band, yet systematic cross-band transfer experiments remain scarce in the current literature. Polarization determines sensitivity to target geometry and structural composition. The cross-polarized channels (HV/VH) are more sensitive to volume-scattering components than co-polarized channels, and training solely on dual-pol Sentinel-1 imagery limits the model's capacity to exploit full-polarimetric information available from sensors such as GF-3. Spatial resolution and incidence angle jointly govern the severity of geometric distortions, including layover and foreshortening, and modulate the radar cross-section of identical targets [17]. These dependencies imply that a pre-training corpus lacking diversity along even one imaging dimension may cause the model to overfit to a narrow operating envelope.

As evidenced by Table 4, many existing pre-training corpora are constructed from a limited set of sensors or sensor families, often with fixed spatial resolution and constrained acquisition configurations. Recent multi-source efforts such as SAR-1M [50] and SARDet-180K [48], which aggregate data across multiple platforms, bands, and polarizations, represent a necessary step toward enabling models to learn acquisition-invariant representations rather than sensor-specific shortcuts.

### 7.2 Evaluation benchmarks: from perception to understanding

Because foundation models target universal representation learning, the conventional practice of random in-distribution splitting tends to overestimate generalization and is insufficient for revealing the true capability boundary of a model. Evaluation protocols have therefore evolved toward more demanding settings that probe different facets of robustness. Out-of-distribution (OOD) and cross-dataset protocols test whether a model captures intrinsic target signatures rather than dataset-specific contextual correlations, by measuring performance across geographically and climatically disjoint regions. Cross-sensor evaluation examines robustness to variations in resolution, noise floor, and system transfer function, for example by transferring representations learned on Sentinel-1 to TerraSAR-X imagery. Cross-geometry evaluation, which assesses performance under varying incidence angles or across ascending and descending orbit passes, has become a critical protocol for verifying whether a model encodes viewpoint-invariant scattering structures. These increasingly rigorous protocols provide the methodological basis for the benchmark taxonomy adopted below. Considering the architectural differences and capability boundaries of existing models, we organize the discussion into three categories: visual perception benchmarks for SAR visual foundation models, semantic cognition and reasoning benchmarks for SAR multimodal foundation models, and generative modeling benchmarks for SAR generative foundation models.

#### 7.2.1 *Benchmarks for SAR visual foundation models*

For foundation models containing visual encoders, the core of the evaluation lies in verifying whether the extracted features exhibit strong robustness to the physical characteristics of SAR. The representation quality is measured across three dimensions: general localization, fine-grained discrimination, and adaptability to complex scenarios. As a fundamental task in visual perception, general object detection first evaluates the model's ability to separate foreground from background in complex scenes. The SSDD [151], SAR-Ship-Det [155], and AIR-SARShip [157] series establish fundamental standards for maritime target detection. For land and aerial scenarios, SAR-Aircraft [173] and SADD [169] are used to evaluate the model's performance in capturing multi-scale targets. Furthermore, large-scale datasets such as LS-SSDD [158] and MSAR [168] can be utilized to test the model's localization robustness in expansive scenes.

Building on this, the evaluation delves deeper into fine-grained discrimination, requiring the model to possess the capability for deep feature disentanglement. OpenSARShip 2.0 [153] and FAIR-CSAR [177] provide type-level fine-grained ground truth for assessing the model's limits in distinguishing similar scattering features. HRSID [159] and SRSDD-v1.0 [163] set benchmarks for berthed target separation and rotated bounding box detection, respectively, testing the model's fine-grained perception of geometric contours.

Finally, to evaluate the model's adaptability under non-ideal imaging conditions, xView3-SAR [162] introduces the dark vessel detection task. Along with SIVED [175] (vehicle detection with rotatable bounding boxes) and OGSOD [174] (SAR object detection), they collectively expand the perception boundary of visual foundation models for complex and high-value targets.

#### 7.2.2 *Benchmarks for SAR multimodal foundation models*

For models incorporating a textual modality, the evaluation focus shifts from purely visual recognition to cross-modal alignment and semantic reasoning. Benchmarking in this domain is no longer restricted to single tasks;

instead, retrieval, zero-shot perception, and logical question-answering are jointly employed to comprehensively assess the model's depth of understanding of SAR semantic concepts.

At the fundamental level of cross-modal alignment, the community primarily relies on the evaluation splits of SARCLIP-1M [19], SAR-TEXT [95], and FSAR-Cap [180] as the mainstream benchmarks. These datasets provide diverse image-text pairs covering rich target variations, and are commonly evaluated using retrieval metrics such as recall to directly quantify the alignment between visual features and the natural-language semantic space.

Building upon such alignment capabilities, evaluation further extends to zero-shot and open-vocabulary perception. Traditional detection datasets such as FUSAR-Ship [161] and RSDD-SAR [170] provide downstream test beds for prompt-based SAR perception when textual category prompts and VLM evaluation protocols are defined.

Furthermore, SARChat-2M [20] provides a large-scale benchmark for SAR-oriented vision-language interpretation. It supports evaluation of whether VLMs can connect SAR visual evidence with textual responses in dialogue-style interpretation tasks.

#### 7.2.3 *Benchmarks for SAR generative foundation models*

With the shift from early GAN-based approaches to more capable diffusion models, the evaluation focus for SAR generative models has moved beyond mere visual realism toward physical consistency and downstream usability. We categorize the existing evaluation benchmarks into three core domains.

**Image synthesis.** This represents the core application scenario of SAR generative models, aiming to address the long-standing challenges of data scarcity and annotation difficulty in SAR ATR and detection tasks. The evaluation focus lies in establishing an integrated generation-labeling pipeline. Within this paradigm, SSDD [151] and HRSID [159] are not only benchmarks for detection tasks, but are also repurposed as downstream validation platforms for generative models. Researchers (e.g., Ship-Go [143]) leverage them to assess whether the generated image-mask pairs can significantly improve detector average precision (AP) on real-world data, thereby validating the effectiveness of synthetic data. Meanwhile, for discrete target generation, MSTAR [150] remains the de facto benchmark for vehicle target synthesis. It is commonly used to evaluate the performance gain of generated samples in few-shot classification scenarios, as well as to compute FID scores for quantifying feature-space distributional similarity between generated and real samples.

Despite the demonstrated effectiveness of generative models in augmenting long-tailed categories, a fundamental physics-consistency gap persists between synthetic and real SAR data. Current diffusion-based generators inherit architectures and noise schedules designed for natural images, where the corruption process is modeled as additive Gaussian noise. SAR speckle, however, is inherently multiplicative and coherent [16]. In homogeneous regions, single-look amplitude may follow a Rayleigh distribution and multilook intensity a Gamma distribution, but in heterogeneous or textured scenes the statistics shift toward heavy-tailed models such as K-distribution and $\mathcal{G}^0$-distribution [181]. Generators that do not account for this distinction may produce visually plausible outputs whose underlying intensity statistics deviate from those of real acquisitions. Furthermore, complex scattering phenomena in real SAR imagery, including multi-bounce interactions in urban canyons and corner-reflector responses from man-made structures, encode fine-grained physical information that current generators tend to reproduce as texture rather than as geometrically consistent scattering signatures [26]. This discrepancy is most pronounced in dense urban areas and rough sea surfaces, where strong scatterers and heterogeneous clutter co-exist. In practice, models trained predominantly on synthetic data exhibit noticeable performance degradation when deployed in such scenes. Addressing this gap requires recognizing a fundamental limitation of current evaluation metrics. FID computes distributional distance in an ImageNet feature space that encodes texture and color statistics with no relevance to microwave scattering; a generated image can score favorably while its amplitude statistics are inconsistent with the expected speckle distribution for the depicted terrain and look number. CLIP Score measures semantic alignment between image and text but cannot assess whether the underlying scattering patterns are physically plausible. Current generative benchmarks therefore evaluate visual and semantic quality but leave radiometric and statistical fidelity largely unmeasured. Developing physics-grounded alternatives is an important open problem, which we discuss in Section 9.

**Image restoration and enhancement.** To address the inherent issues of speckle noise and low spatial resolution in SAR imaging, this category of benchmarks focuses on evaluating a model's ability in signal restoration and fine-detail enhancement. SAR restoration and enhancement can be evaluated using task-specific SAR data sources and degradation settings, including synthetic speckle/noisy-clean pairs, real noisy SAR observations, downsampled high-resolution SAR images for super-resolution, no-reference image-quality metrics, and downstream recognition or detection performance. Existing despeckling and restoration studies commonly combine simulated degradations with real SAR observations to assess both quantitative fidelity and practical utility.

**Text-guided generation and controllable synthesis.** With the advancement of generative AI technologies, generating SAR images with specific attributes from natural-language prompts has become a research frontier, shifting the evaluation focus from unconditional generation to semantically controllable generation. In this context, SARCLIP-1M [19] and SAR-TEXT [95] provide large-scale SAR image-text pairs for semantic conditioning and alignment evaluation. When paired with trained generators, evaluation in this domain can jointly consider physical realism, downstream utility, and semantic consistency. Researchers may compute CLIP Score [123] using pretrained SAR vision-language models to quantify the alignment between generated imagery and textual instructions, including target orientation and environmental context.

The benchmark landscape described above reveals that evaluation settings in SAR vary substantially across tasks, sensors, and modalities. Visual foundation models are typically assessed under cross-resolution and cross-sensor conditions (e.g., transferring between Sentinel-1 and GF-3), where the model must generalize across different imaging geometries and noise levels. Multimodal benchmarks introduce cross-modal alignment requirements, demanding that representations remain consistent between SAR and textual or optical inputs. Generative benchmarks further impose physical fidelity constraints, requiring synthesized images to preserve scattering statistics rather than merely achieve perceptual similarity. These divergent evaluation criteria directly shape the selection of downstream adaptation strategies. Tasks evaluated under narrow, sensor-specific settings often favor full fine-tuning to achieve maximal feature reshaping, whereas benchmarks that emphasize cross-domain generalization place a premium on preserving pretrained representations, making parameter-efficient methods or prompt-based inference more appropriate. This connection between evaluation protocol and adaptation design motivates the systematic discussion of adaptation techniques that follows.

Dataset value is determined less by raw sample count than by diversity across sensors, bands, polarizations, geometries, and scenes. Evaluation protocols should emphasize cross-sensor, cross-geometry, and physics-aware tests rather than only random in-distribution splits.

## 8 Adaptation techniques

The pre-training process of SAR foundation models is, in essence, an information compression and knowledge-encoding procedure over massive microwave scattering signals, enabling the models to internalize general scattering mechanisms and manifold structures of ground objects. However, completing pre-training merely indicates that the model has acquired latent, general-purpose cognition; to translate this into practical task capability for addressing specific remote sensing problems (e.g., ship detection, land-cover classification), it is necessary to bridge the gap from general representations to task-specific decision-making.

### 8.1 Fine-tuning paradigms

This section provides an in-depth discussion of downstream adaptation and inference paradigms for SAR FMs. As illustrated in Figure 13, these adaptation techniques do not exist in isolation; rather, they form an integrated spectrum ranging from inference-time interventions with low computational cost to parameter reconfiguration with high computational cost. We decompose this process along two dimensions: fine-tuning, which examines how to achieve parameter reconstruction and alignment of model parameters in the SAR data domain through either full fine-tuning or PEFT; and inference paradigms, which investigates how, under frozen parameters, prompt engineering and feature mining can be leveraged to elicit the model's reasoning potential in zero-shot or few-shot settings.

The emergence of foundation models has fundamentally reshaped the technical paradigm of SAR image interpretation, driving the field's transition from training from scratch [182,183] to a pre-train-then-adapt workflow. As the core component of this paradigm, fine-tuning essentially leverages a small amount of downstream labeled data to effectively transfer and transform the general features learned during pre-training into task-specific capabilities. However, in the SAR domain, this process faces a severe plasticity-stability dilemma. On the one hand, the high sensitivity of SAR imaging demands that the model exhibit strong plasticity. Unlike optical images, the visual appearance of SAR targets varies dramatically with observation conditions (e.g., incidence angle, frequency band). A pretrained model acquires broad general features, whereas downstream tasks often focus on specific imaging geometries or sensor parameters. This pronounced discrepancy in feature distributions forces the model to substantially adjust its parameters in order to reconstruct general representations into precise representations tailored to the target task. On the other hand, the practical constraint of limited samples can easily undermine the model's stability. Downstream SAR tasks typically lack large-scale labeled datasets. Under data scarcity, aggressively updating all model parameters is highly likely to cause the model to discard the general, structured knowledge acquired during pre-training in order to fit the details (or even noise) of the current samples, thereby triggering catastrophic

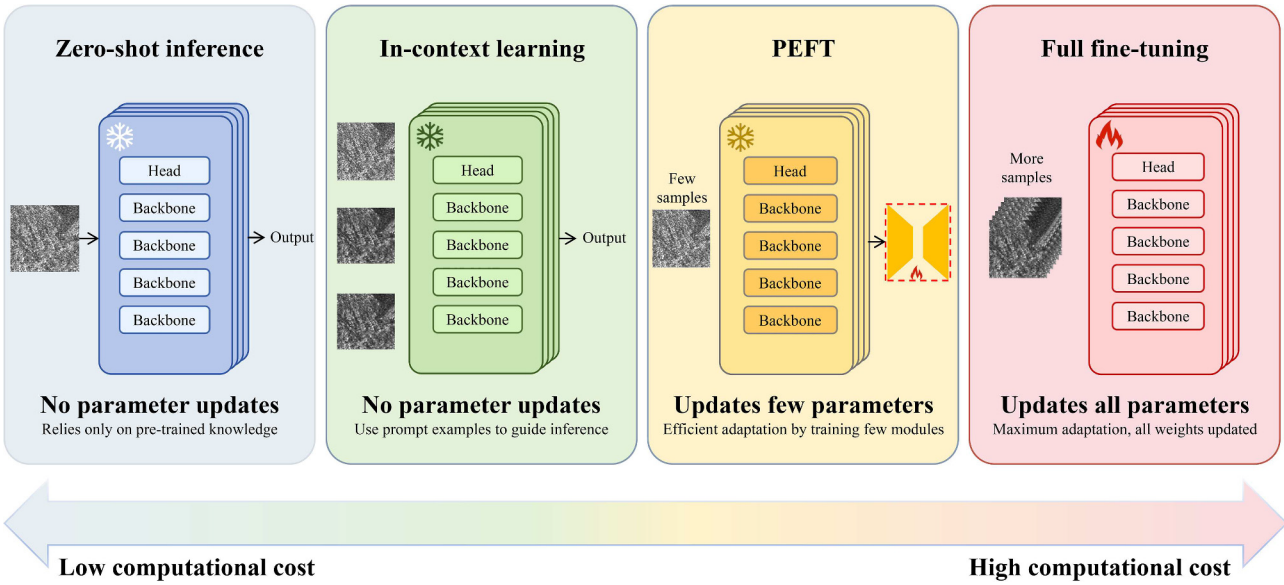



**Figure 13** (Color online) Comparison of four large-model adaptation paradigms, covering zero-shot, few-shot in-context learning, PEFT, and full fine-tuning, revealing the trade-off between accuracy and computational cost.

forgetting. Therefore, the choice of a fine-tuning strategy is essentially about seeking an optimal balance between adapting to task-specific discrepancies and preserving the general pretrained knowledge [184].

### 8.1.1 *Full fine-tuning*

Full fine-tuning (FFT) uses the pretrained model parameters as initialization and performs complete backpropagation updates for all layers. The key advantage of FFT lies in its deep feature reshaping capability: when dealing with SAR images from different domains and cross-modal image sources, adjusting only the top-layer parameters is often insufficient to accommodate the heterogeneity of low-level features. FFT enables the model to bridge this gap, thoroughly remapping the pretrained general representations into downstream-task visual representations, and, in theory, can approach the performance upper bound for a specific task [184]. The study by Pu et al. [185] validates this advantage: they abandoned optical weights and, by conducting MAE full self-supervised pre-training on SARDet-100K followed by fine-tuning, significantly improved detection accuracy. SAR-HUB [186] further demonstrates that, with sufficient data support, full-parameter updates can maximize the exploitation of dataset-specific information in SAR data, and its feature extraction capability surpasses methods that freeze the backbone network.

However, FFT is extremely sensitive to data distributions. Performing aggressive fine-tuning directly on heterogeneous and scarce SAR data can easily lead to overfitting or damage the pretrained features. To address this challenge, researchers have explored progressive transfer strategies, which aim to smooth the loss landscape of optimization by constructing intermediate domains. Liu et al. [187] designed a curriculum learning route of natural images $\rightarrow$ simulated SAR $\rightarrow$ real SAR, building a smooth transition bridge across heterogeneous modalities and effectively avoiding gradient oscillations. Another approach is to expand the data to suppress overfitting. Given FFT's strong dependence on the number of labeled samples, generative augmentation has become a key technique. The Ship-Go diffusion model proposed by Zhang et al. [143] fills the long-tail distribution of training data by generating diverse SAR ship instances, enhancing the robustness of full fine-tuning from the data source. This is consistent with the early idea of Wang et al. [47], who leveraged data augmentation to expand SAR datasets: both support the update demands of large parameter counts by enriching the data manifold.

Although the above strategies substantially improve the robustness of FFT, its non-negligible computational cost and risk of overfitting have driven research to increasingly shift toward more lightweight solutions. The traditional backbone freezing strategy [188], while extremely efficient, often comes at the expense of the achievable performance upper bound. This dilemma between full updates and complete freezing has directly given rise to the emergence of PEFT.

### 8.1.2 *Parameter-efficient fine-tuning (PEFT)*

PEFT represents a new paradigm for adapting foundation models [189]. Its core logic is to freeze the vast majority of parameters in the pretrained backbone and inject SAR-domain knowledge by updating or adding only a small

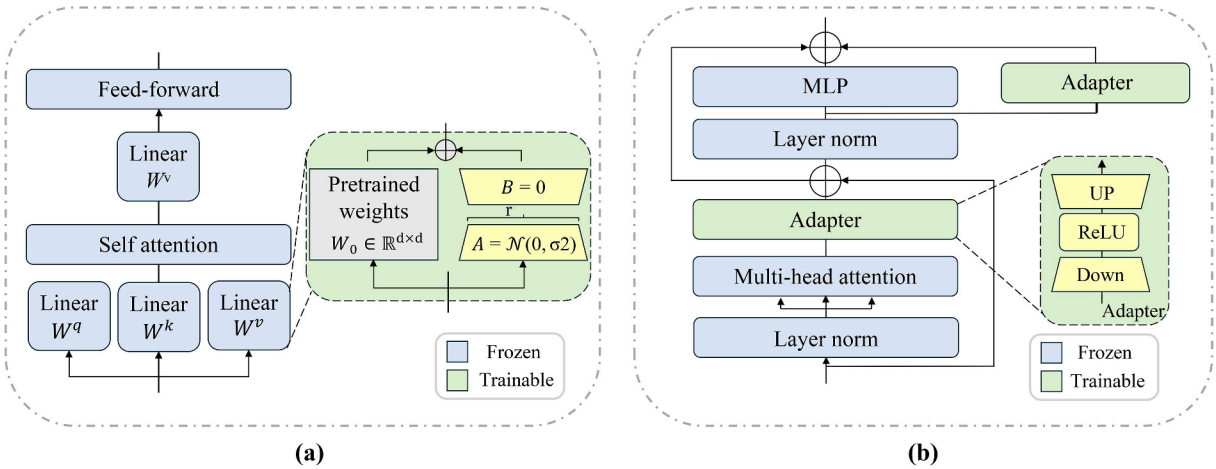



**Figure 14** (Color online) Schematic diagram of the principles of two fine-tuning methods. (a) LoRA fine-tuning diagram; (b) adapter fine-tuning diagram.

number of parameters. The essential advantage of this approach lies in the fact that it treats the SAR-specific physical scattering mechanisms as a form of incremental knowledge and injects it into the general model in a targeted manner. This architecture decouples general visual capabilities from SAR physical characteristics, thereby preserving the foundation model's generalization ability while enabling deep adaptation to SAR tasks at a low parameter cost.

**Low-rank adaptation (LoRA).** As illustrated in Figure 14, LoRA [190] is based on the assumption that the model's update has a low intrinsic rank. By optimizing low-rank matrices in a bypass branch, it implicitly simulates full-parameter updates, and has become a mainstream choice for balancing computational efficiency and adaptation performance.

Given a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, LoRA freezes $W_0$ and introduces low-rank matrices $A \in \mathbb{R}^{d\times r}$ and $B \in \mathbb{R}^{r\times k}$ (where $r \ll d$) to approximate the parameter update $\Delta W$. The forward propagation is formalized as

$$h = W_0 x + \Delta W x = W_0 x + ABx. \tag{5}$$

This bypass updating mechanism reduces the number of trainable parameters from $d \times k$ to $r(d + k)$, greatly decreasing GPU memory usage and laying the foundation for deployment on edge devices. For example, SaRA proposed by Hu et al. [191] further compresses LoRA's parameter space via sparsification strategies, which is of significant engineering value for deploying high-fidelity generative models on satellite and airborne edge platforms.

At the level of multimodal semantic understanding, LoRA demonstrates an efficient compatibility between general language priors and SAR visual features. Experiments by Wei et al. [192], which apply LoRA fine-tuning to the linear layers of VLMs such as LLaVA-1.5, show that updating only a very small number of parameters enables the model to accurately comprehend complex scene semantics in SAR images. This implies that, although SAR and optical images differ greatly in visual appearance, they share similar topological structures in the high-level semantic space. The practice by Tai et al. [193], which performs ASNARO-2 LoRA fine-tuning on a pretrained ViT, further confirms that such low-rank updates are sufficient to mitigate data heterogeneity introduced by different sensors.

Generative tasks expose a clearer semantic-physical trade-off for LoRA. General diffusion backbones, such as Stable Diffusion, encode optical-domain priors for smooth textures. LoRA can efficiently adjust attention projections, but these low-rank updates may not fully reshape the pixel-generation distribution needed for SAR.

Two implications follow. First, LoRA can preserve textual semantic consistency, as shown by text-guided controllable SAR generation in Wu et al. [194]. Second, it may still miss SAR-specific high-frequency details, including fine edges, speckle statistics, and scattering points. A practical compromise is the hybrid strategy of Debuysère et al. [149]: full fine-tuning is applied to the U-Net for geometric realism, while LoRA is used on the text encoder to preserve semantic controllability.

**Adapter-based mechanisms.** Unlike LoRA's bypass updates, adapter-based methods explicitly insert modules into the network architecture to adjust the feature flow. A typical representative is ClassWise-SAM-Adapter proposed by Pu et al. [93] for the Vision Transformer architecture. This method inserts a simple and efficient adapter within a single module while freezing all original parameters during training. Such a design not only greatly reduces the training cost, but more importantly, constructs a decoupled architecture of a general visual backbone coupled with an SAR physical adapter, enabling the model to switch rapidly to accommodate different

**Table 5** Comparative summary of adaptation strategy families for SAR foundation models. Parameter ratio denotes the proportion of trainable parameters relative to the total model size. GPU memory is reported relative to FFT. Forgetting risk indicates the likelihood of catastrophic forgetting of pretrained knowledge. The reported ranges and data-scale regimes are illustrative and implementation-dependent.

| Adaptation strategy | Parameter ratio | GPU memory | Forgetting risk | Suitable data scale |
|---|---|---|---|---|
| FFT | 100% | 1× (baseline) | High | Large (>10k samples) |
| LoRA | <1%–5% | ∼0.3×–0.5× | Low | Medium (1k–10k samples) |
| Adapter | ∼2%–5% | ∼0.4×–0.6× | Low | Medium (1k–10k samples) |
| Prompt tuning/visual prompting | <1% | ∼0.2×–0.4× | Very low | Small (<1k samples) |
| Linear probing/few-shot | <0.1% (head only) | ∼0.1×–0.2× | Negligible | Very small (<100 samples) |

SAR sensors or task requirements and demonstrating strong generalization potential. RingMo-SAM [94] and Hu et al. [195] adopt a similar strategy: by introducing a scattering prompt adapter, they successfully transform SAR high-frequency scattering features into visual prompts that SAM can interpret, achieving cross-domain SAR object detection and segmentation without compromising SAM's strong segmentation capability.

To provide a systematic comparison of the adaptation strategies discussed above, Table 5 summarizes the key trade-offs across five representative adaptation families. The comparison dimensions include the ratio of trainable parameters to total model parameters, the GPU memory footprint relative to full fine-tuning, the risk of catastrophic forgetting of pretrained knowledge, and the data-scale regime in which each strategy is most effective. It should be noted that the specific values of parameter ratios and memory footprints depend on the backbone architecture and implementation details; the ranges reported here reflect typical configurations observed in the SAR foundation model literature reviewed in this survey.

## 8.2 Inference paradigms: prompting and probing

Unlike fine-tuning strategies, which actively update model parameters, inference-time adaptation [196] aims to keep the pretrained model weights frozen and achieve rapid adaptation to downstream tasks by designing adaptive input interaction interfaces (e.g., prompt engineering) or by leveraging the strong generalization ability of frozen features. This paradigm has particular value in the SAR domain: it not only fundamentally avoids the risk of catastrophic forgetting caused by full training under small-sample conditions, but also enables a single foundation model to serve diverse SAR interpretation tasks (e.g., classification, detection, and segmentation) at extremely low marginal cost, thereby truly unleashing the general-purpose potential of foundation models.

### 8.2.1 *Alignment-based zero-shot prompting*

The core objective of zero-shot learning is to endow a model with the ability to recognize unseen classes that are not observed during training. Its basic paradigm is to use labeled data from seen classes, together with semantic auxiliary information such as attributes or word vectors, to construct a mapping between the visual features and the semantic space, thereby enabling inference on unseen classes [197].

In terms of text prompting, SAR-CLIP [19] established an early benchmark. Through contrastive pre-training on millions of SAR-text pairs, the model learns a deep correspondence between visual features and natural-language descriptions, enabling zero-shot classification of unseen targets via natural-language prompts for a given SAR image (see also Section 5 for its architectural details). However, it should be noted that the non-intuitive nature of SAR imagery makes natural language difficult to precisely describe certain complex scattering characteristics (e.g., dihedral reflections and geometric distortions), which limits the upper bound of alignment. To break this physics-semantics bottleneck, VLF-SAR [198] introduces a frequency-band embedding module (FEM), incorporating frequency-domain priors into multimodal alignment to enhance the model's understanding of physical concepts, thereby achieving deeper alignment between natural language and the physical concepts in SAR images. As shown in Figure 15, this process typically consists of three key steps: first, use the frozen encoders learned from large-scale pre-training to extract the visual features of the input SAR image and the text features of candidate categories, respectively; second, compute the similarity matrix between the two in a shared latent space; finally, retrieve the text prompt with the highest similarity score as the prediction. The advantage of this mechanism is that it completely avoids parameter updates, allowing the model to instantly handle long-tail targets that were not seen before, such as ship wakes. In addition, to address the fundamental challenge of lacking high-quality image-text paired data at the source, Wang et al. [142] proposed a simulation-based reverse-alignment paradigm, which uses diffusion models to transform rich optical semantics into 3D SAR simulation data, thereby enhancing the quality of vision-semantics alignment from the data source.

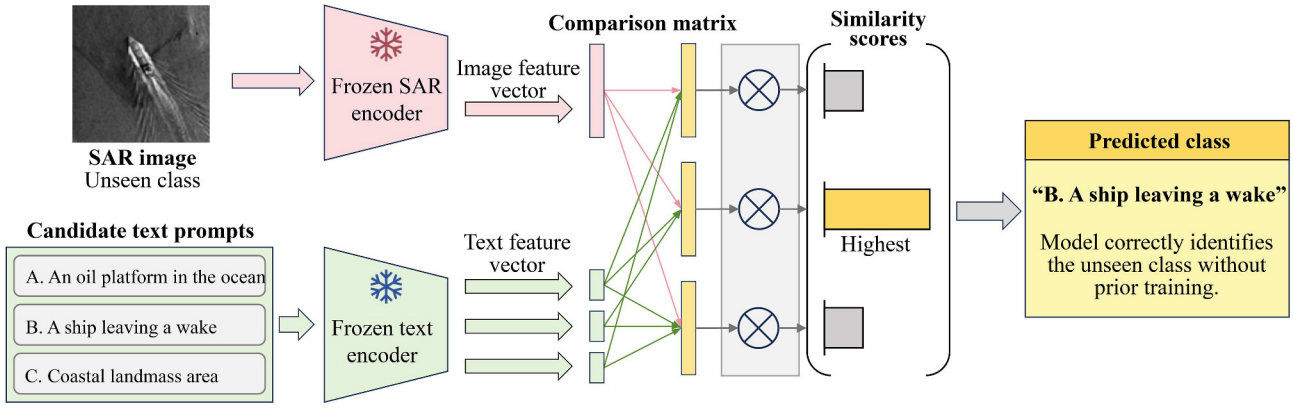



**Figure 15** (Color online) Schematic of the zero-shot inference pipeline based on vision-language alignment. This paradigm requires no parameter updates and directly leverages the frozen SAR encoder learned during pre-training.

In the dimension of visual prompting, for adapting large segmentation models (e.g., SAM) to the SAR domain, prompt learning enables adaptation through learned prompts while keeping the pretrained backbone frozen. RSPrompter proposed by Chen et al. [43] draws inspiration from prompt learning and designs a learnable prompt-generation network. Instead of relying on manually crafted point or box prompts, this method learns to generate soft prompts tailored for SAM, enabling the frozen SAM to understand semantic boundaries in SAR images and thereby achieving strong instance-segmentation performance on datasets such as SSDD. The significance of this finding is that it demonstrates: for general-purpose visual foundation models, designing appropriate visual prompts to bridge the gap between the SAR domain and the natural-vision domain is a more robust and efficient strategy than forcibly fine-tuning model parameters.

Beyond the general prompt-learning paradigm discussed above, a particularly noteworthy direction in the SAR domain is the injection of physics-informed prompts that encode domain-specific scattering knowledge into frozen foundation backbones. Unlike conventional text or visual prompts that operate purely in the semantic or pixel space, SAR-specific prompting strategies exploit the unique physical properties of microwave scattering to construct prompts with explicit physical meaning. A representative example is the scattering prompt adapter introduced by Hu et al. [195], which extracts high-frequency scattering features from SAR images, including edge responses and point-scatterer signatures characteristic of dihedral and trihedral reflections, and transforms them into prompt tokens that are injected into the intermediate layers of a frozen SAM encoder. This mechanism effectively communicates SAR-specific geometric and radiometric information to a foundation model pretrained exclusively on optical imagery, enabling it to correctly delineate targets whose boundaries are defined by scattering discontinuities rather than photometric gradients. Similarly, the FEM in VLF-SAR [198], while primarily designed for vision-language alignment, can also be interpreted as a form of frequency-domain prompting: it encodes sub-band spectral signatures as auxiliary tokens that are fused with the visual features before cross-modal alignment, thereby injecting frequency-dependent scattering priors into the representation space.

These SAR-specific prompting cases share a common design principle: rather than adapting the model's internal representations through parameter updates, they reformulate the physical priors of SAR imaging (scattering geometry, frequency-domain statistics, and polarimetric signatures) as structured input signals that guide the frozen model's inference process. This paradigm is especially promising for multi-sensor SAR scenarios, where the same foundation model may need to process data from sensors operating at different frequency bands (e.g., X-band vs. L-band) with markedly different scattering characteristics. By switching only the physics-informed prompt module while keeping the backbone frozen, one can efficiently accommodate sensor-specific scattering behaviors without retraining the entire model.

### 8.2.2 *Representation-based few-shot probing*

Unlike prompt-based zero-shot inference that requires absolutely no parameter updates, few-shot learning (FSL) via probing represents an extremely lightweight parameter-adaptation paradigm. It aims to build a classifier with strong generalization ability using only a small number of labeled samples per class [199]. Its core idea is that if a foundation model has already learned sufficiently robust, general-purpose SAR representations, high-performance recognition can be achieved by strictly freezing the entire pretrained backbone and only training a newly initialized classification head (e.g., a linear probe) or constructing a metric space using a very small support set.

Linear probing is a standard paradigm for evaluating the quality of foundation-model representations. Inkawhich et al. [200] conducted an in-depth investigation into the mechanisms of self-supervised contrastive learning frameworks (SimCLR [96]) on SAR data. Their experiments show that, by using a frozen feature extractor alongside a trainable linear classification layer, the few-shot performance on the MSTAR dataset can significantly outperform supervised models trained from scratch. Building on this, Liao et al. [59] pointed out that the high similarity of SAR images makes conventional contrastive learning struggle to distinguish fine-grained features. Their proposed LDCL introduces a low-confidence strategy and a group-instance discrimination mechanism to specifically mine confusing hard samples. Experimental results demonstrate that LDCL exhibits stronger feature discriminability than SimCLR under linear evaluation, indicating that optimizing the topological structure of the feature space is the fundamental way to improve linear probing performance.

Metric learning is another mainstream pathway for few-shot adaptation. Shang et al. [201] improved convolutional neural networks by introducing an information-recording mechanism to store the spatial features of labeled samples, and propagated pseudo-labels to unlabeled samples by measuring similarity in the feature space (e.g., Euclidean distance or cosine similarity). In the era of foundation models, this approach has evolved into computing prototypes using pretrained features. The survey by Abdollahzadeh et al. [202] further points out that combining generative modeling to expand the feature space (feature hallucination) is an important future direction for improving the robustness of few-shot inference.

Overall, from linear-space optimization based on SimCLR to metric-space reconstruction, inference-time adaptation paradigms are gradually shifting their focus from parameter fine-tuning to the intrinsic structure of the feature manifold. This lightweight adaptation mode provides theoretical support for deploying SAR foundation models on edge devices with constrained computational resources.

#### 8.2.3 *Route-selection guideline*

The preceding subsections show that SAR foundation model adaptation involves a spectrum of strategies, ranging from full parameter updates to frozen-backbone inference. These strategies differ in adaptation capacity, data requirement, computational cost, and the risk of forgetting pretrained representations. Therefore, the choice of adaptation strategy should be determined by four factors: the domain gap between pre-training and target data, the amount of labeled data, the available compute budget, and the downstream task type.

When the domain gap is large and sufficient labeled data are available, full fine-tuning is generally more effective because it allows deeper feature reshaping. When labeled data or compute resources are limited, PEFT methods such as LoRA and adapters are usually more suitable, as they preserve pretrained knowledge while still providing task-specific adaptation. In low-resource or open-vocabulary settings, prompting and linear probing are often preferable because they require minimal parameter updates and reduce the risk of overfitting. For deployment scenarios involving multiple sensors or frequent task switching, modular adapters or prompt-based designs offer greater flexibility than retraining the full model. Generative tasks impose an additional constraint, since physical fidelity and semantic controllability may need different adaptation depths in different components, making hybrid strategies a practical choice. At the system level, these deployment-oriented choices are also related to the broader trend of communication-computation co-design, where emerging AI-native communication infrastructures may support distributed training, inference, and model adaptation across edge and cloud nodes [203].

Full fine-tuning is appropriate when domain shift is large and labeled data are sufficient, whereas PEFT, prompting, and probing are safer under scarce labels or constrained compute. Adaptation should be chosen as a deployment decision rather than as a fixed recipe.

## 9 Future directions

Compared with the flourishing foundation-model ecosystem in computer vision and natural language processing, research on SAR foundation models, although having achieved notable progress, still remains at an initial, exploratory stage. Considering the distinctive physical properties of SAR data and its critical value in fields such as national defense and disaster monitoring, this area has enormous development potential and application value. Below, we outline several promising research directions.

(1) **Construction of high-quality, large-scale, and diverse datasets.** The first bottleneck is the lack of SAR datasets that satisfy the scale, diversity, and physical reliability required for foundation-model pre-training. Future dataset construction should prioritize three aspects.

• **Standardized processing.** Noise suppression, radiometric calibration, and geometric correction should be applied consistently, with key attributes such as land-cover category and target type recorded in a structured form.

• **Physically informed annotation.** Beyond class labels, CoT annotations can describe scattering cues such as corner reflection, layover, and shadowing, together with the imaging-geometry reasoning behind the final interpretation.

• **Diverse acquisition coverage.** Data should span spaceborne and airborne platforms, L/C/X frequency bands, multiple polarizations, varied resolutions, long time series, and heterogeneous paired modalities such as optical imagery and text.

Such resources would support models that learn not only semantic categories, but also radar-specific physical reasoning and cross-sensor generalization.

(2) **Development of multimodal, strongly generalizable, and multifunctionally integrated SAR agents.** With the evolution of artificial general intelligence, multimodal fusion, strong generalization, and multifunctional integration have become core trends in the development of foundation models. Future SAR foundation models may evolve toward more interactive agents that connect perception, reasoning, and deployment. RingMoAgent [127] demonstrates a remote-sensing multimodal reasoning route by using RS-VL3M-scale vision-language data across optical, SAR, and infrared imagery from satellite and UAV platforms, together with modality-adaptive embeddings and task-specific decoding for perception and long-horizon spatial reasoning. REMSA [128] adds a deployment-oriented direction by selecting foundation models under task and resource constraints. For multimodal fusion, it is necessary to explore deep fusion mechanisms between SAR data and multi-source modalities such as optical remote-sensing data, infrared data, geospatial data, and text descriptions. Through cross-modal alignment and joint representation learning, the texture advantages of optical imagery can alleviate the difficulty of visual interpretation in SAR, while SAR's penetration advantages can compensate for the limitations of optical imagery, enabling holistic perception of complex scenes. Meanwhile, to build strong generalization ability, it is essential to overcome the inherent domain-shift challenges of SAR data, enhancing the model's adaptability to unseen scenarios and tasks and reducing dependence on scenario-specific data. On this basis, future systems can move beyond single-task architectures toward integrated SAR interpretation workflows that connect target detection, land-cover classification, change detection, image segmentation, parameter inversion, and scene understanding.

(3) **Research on interpretability and trustworthiness.** SAR foundation models are mostly black-box models, and the lack of interpretability in their decision processes severely limits their deep application in critical and sensitive areas such as national defense security and disaster emergency response. Therefore, improving interpretability and trustworthiness is an important future research direction. This requires introducing techniques such as attention visualization, causal analysis, and inverse reasoning to analyze the relationship between SAR scattering features extracted by the model and their physical meanings. It also requires incorporating physical prior knowledge, such as radar electromagnetic scattering mechanisms, into deep learning models to build physics-guided foundation models. Notably, CoT methods can serve as a core technical pathway within this research framework: by stepwise decomposing the model's decision chain, the relationship between extracted scattering features and physical meanings can be clearly traced, allowing the model's decision process to move from being knowable in outcome to traceable in process. Furthermore, establishing a comprehensive uncertainty assessment mechanism for model predictions is essential. When the model faces ambiguous targets or complex backgrounds, it should be able to output credible confidence scores, thereby ensuring the safety and reliability of the system in real-world applications.

(4) **Robustness enhancement for extreme scenarios.** SAR technology is often applied in extreme scenarios such as typhoons, floods, earthquakes, and battlefields. In these environments, SAR data often face severe challenges including strong speckle noise, nonlinear deformation, target incompleteness, complex electromagnetic backgrounds, and potential adversarial interference, leading to significant performance degradation of existing foundation models. Future research must focus on improving model robustness under extreme scenarios, developing robust feature extraction algorithms for low signal-to-noise ratio and geometrically distorted environments, in order to enhance recognition and analysis capabilities under harsh imaging conditions. In special application contexts such as the military, research on adversarial-attack defense techniques and few-shot learning methods is also crucial. This will ensure that the model can still maintain stable performance output under data incompleteness or interference, meeting the stringent requirements of practical operations and emergency response.

(5) **Establishment of physics-aware evaluation protocols.** Current evaluation of SAR generative models relies on metrics such as FID and CLIP Score, which measure perceptual similarity or semantic alignment but are agnostic to SAR imaging physics. A generated image may achieve a favorable FID score while its intensity statistics deviate from distributions expected under a given number of looks and terrain type. Future benchmarks should incorporate three physics-grounded dimensions. The first is scattering-statistics testing: comparing amplitude histograms of generated imagery against theoretical models such as Rayleigh, K, or $\mathcal{G}^0$ distributions using goodness-of-fit tests to quantify distributional fidelity. The second is spectral consistency: measuring divergence between the

power spectral densities of generated and real patches to verify faithful reproduction of point-scatterer responses and edge structures. The third is radiometric fidelity: verifying that calibrated backscatter coefficients ($\sigma^0$) of generated images fall within physically plausible ranges for the depicted land-cover classes. Beyond generative evaluation, similar criteria can benefit visual and multimodal models, for example by testing whether learned representations preserve the known relationship between backscatter intensity and incidence angle. Standardized protocols along these dimensions would equip the community with rigorous tools for distinguishing visual plausibility from genuine physical consistency.

(6) **Understanding scaling behavior under SAR data redundancy.** A fundamental open question is whether SAR foundation models follow scaling behavior similar to that observed in NLP and optical vision, where performance improves predictably with increasing model size, data volume, and compute. At present, the answer remains unclear. Unlike natural-image corpora, SAR datasets often contain substantial redundancy, because adjacent patches, repeated acquisitions, and homogeneous backgrounds can contribute many visually similar samples while adding limited new information about scattering diversity. This issue is particularly relevant in domains such as ocean monitoring, where large amounts of data may be dominated by repetitive clutter. Existing evidence already suggests that scaling in SAR is not a simple matter of increasing sample count. As discussed in Section 4, OceanSAR-1 shows that reducing semantic redundancy through dynamic data curation can be more beneficial than naively enlarging the training set. This observation implies that the effective scaling variable in SAR may not be raw dataset size, but the joint diversity of geography, sensor type, frequency band, polarization, imaging geometry, and scene complexity. Future work should therefore move beyond reporting results on isolated datasets and establish controlled scaling studies that vary model size, training data volume, and compute budget simultaneously. Such studies should also distinguish between nominal data growth and effective information growth, so that the community can determine when larger SAR corpora truly improve generalization and when performance saturates because the added data are redundant.

Future progress will depend on larger but more physically diverse data, stronger multimodal agents, and evaluation protocols that measure trustworthy SAR behavior. The central challenge is to scale models without losing scattering interpretability, robustness, and radiometric consistency.

## 10 Conclusion

This review has surveyed the emerging field of SAR foundation models and organized existing work into a three-paradigm taxonomy covering visual, multimodal, and generative approaches. Beyond cataloguing methods, the preceding sections have identified several evidence-backed findings that merit explicit statement.

A central question in this field is whether SAR-specific pre-training is necessary or whether generic vision foundation models can be directly transferred. The evidence reviewed in Section 4 indicates that the answer is task-dependent. For tasks that rely heavily on SAR scattering characteristics, such as target recognition under extended operating conditions and cross-sensor transfer, generic optical pre-training serves primarily as parameter initialization, and the performance gains critical to SAR emerge from subsequent domain-specific pre-training. SARATR-X [48], for instance, validates this two-stage strategy across five datasets and eight evaluation settings. Conversely, when SAR physical constraints are explicitly injected at the inference stage, generic priors can also be successfully adapted, suggesting that the decisive factor is not whether the backbone was pre-trained on SAR data per se, but whether SAR imaging physics is accounted for at some stage of the pipeline. A unified benchmark that compares these two routes under controlled conditions across multiple tasks and sensors does not yet exist, and establishing one is an important next step.

Regarding SAR-optical complementarity, the evidence reviewed in Section 5 confirms that joint representations yield consistent and measurable gains over SAR-only representations, with improvements of approximately 5–6 mAP reported under low-annotation settings by both CROMA [63] and DeCUR [64]. These gains, however, are not unconditional. Forcing all modality features into a shared subspace suppresses modality-unique information and reduces downstream performance, as demonstrated by the ablation analyses of both models. The practical implication is that fusion architectures should explicitly preserve SAR-specific representations alongside shared cross-modal features, rather than pursuing maximal alignment. This lesson extends to the generative paradigm, where evaluation metrics borrowed from the optical domain fail to capture SAR-specific physical properties, creating a risk that models producing visually plausible but physically inconsistent outputs are judged favorably.

Looking forward, the directions outlined in Section 9, including large-scale multi-sensor dataset construction, physics-aware evaluation protocols, SAR agent engineering, and interpretability grounded in scattering mechanisms, collectively point toward a field that must mature from method proliferation to rigorous, reproducible, and physically

grounded evaluation. Overcoming the challenges of data scarcity and cross-domain heterogeneity while preserving the physical consistency unique to SAR will be essential for realizing the potential of all-weather, day-and-night intelligent Earth observation systems.

**Acknowledgements** This work was supported by National Natural Science Foundation of China (Grant No. 62425115).

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